Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 3

Computational approaches

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch03/code/ch03.R` (hand-written samplers, base R). The JAGS versions are in `ch03/code/ch03_jags.R`; they run automatically on GitHub.

Sai Sridhar · October 20, 2026

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A five-minute primer on the tools of this chapter

Almost every exercise in this chapter uses one of the following ideas. Read this page once; the solutions refer back to it.

Background you need

1. The posterior up to a constant. Bayes' rule says $p(\theta \mid \mathbf{Y}) = \dfrac{f(\mathbf{Y} \mid \theta)\pi(\theta)}{m(\mathbf{Y})}$. The denominator $m(\mathbf{Y})$ does not depend on θ, so we usually write

$$p(\theta \mid \mathbf{Y}) \;\propto\; f(\mathbf{Y} \mid \theta)\,\pi(\theta) \qquad (\text{"likelihood} \times \text{prior"}).$$

We work on the *log scale*, $\log p = \log f + \log \pi + \text{const}$, because products become sums and numbers stay in a safe range.

2. MAP estimate. The *maximum a posteriori* estimate is the peak of the posterior: solve $\frac{\mathrm{d}}{\mathrm{d}\theta} \log p(\theta \mid \mathbf{Y}) = 0$.

3. Bayesian CLT. With enough data the posterior is approximately normal, centred at the MAP, with variance equal to the inverse of the curvature:

$$\theta \mid \mathbf{Y} \;\approx\; \text{Normal}\Big(\hat{\theta}_{MAP},\; \big[-\tfrac{\mathrm{d}^2}{\mathrm{d}\theta^2} \log p(\theta \mid \mathbf{Y})\big|_{\hat{\theta}}\big]^{-1}\Big).$$

4. Full conditional distribution. For a model with several parameters, the full conditional of one parameter is its distribution *given the data and all the other parameters*. Recipe: write the joint "likelihood × prior", keep only the factors that contain the parameter, and recognise the shape.

5. Gibbs sampling. Start somewhere; then repeatedly draw each parameter from its full conditional, using the latest values of the others. After a burn-in period, the draws behave like (correlated) draws from the posterior.

6. Metropolis–Hastings (random walk). When a full conditional has no standard form: propose $\theta^* = \theta + \text{noise}$, compute

$$R = \frac{p(\theta^* \mid \mathbf{Y})}{p(\theta \mid \mathbf{Y})} \quad (\text{only ratios needed — the unknown constant cancels}),$$

accept θ^* with probability $\min(1, R)$, otherwise keep θ. The noise size (step) is tuned so that roughly 30–50% of proposals are accepted.

7. Convergence diagnostics. *Trace plot*: the chain should look like a "hairy caterpillar" with no trend. $\hat{R}$ *(Gelman–Rubin)*: run several chains from different starting points; $\hat{R} \approx 1$ (below 1.01–1.1) means they agree. *Effective sample size (ESS)*: the number of independent draws the correlated chain is worth; aim for at least a few hundred to a thousand.

Exercise 3.1 — Pros and cons of five computational methods

Exercise 1 (exact text from the book)

Give an advantage and a disadvantage of the following methods:

(a) Maximum a posteriori estimation

(b) Numerical integration

(c) Bayesian central limit theorem

(d) Gibbs sampling

(e) Metropolis–Hastings sampling

In plain words

For each way of "computing a posterior", say one thing it does well and one thing it does badly.

Key idea

Ask two questions of every method: *(i) how much of the posterior does it give me* (a single number? the whole distribution?), and *(ii) how does it cope as the number of parameters grows?*

Solution

Method	**Advantage**	**Disadvantage**
(a) MAP	Fast: only an optimisation problem (works for thousands of parameters); no integration needed.	Gives a single point, *no uncertainty*; the mode can be unrepresentative of a skewed or multi-modal posterior.
(b) Numerical integration	Very accurate (essentially exact) posterior means, probabilities and plots for 1–2 parameters; no randomness.	Cost explodes with dimension (a grid of 100 points per parameter needs 100^p evaluations) — impossible beyond 3–4 parameters.
(c) Bayesian CLT	Only needs the MAP and the curvature; gives an instant normal approximation with standard errors.	Only valid for large samples; wrong for skewed posteriors, small n, or parameters near a boundary.
(d) Gibbs sampling	No tuning: each step is an exact draw from a known distribution, and it scales to many parameters.	Needs every full conditional to be a recognisable distribution; mixes very slowly when parameters are strongly correlated.
(e) Metropolis–Hastings	Works for *any* posterior we can evaluate up to a constant (no conjugacy needed).	Must be tuned (step size); poor tuning gives highly correlated draws, and it struggles in high dimensions.

Answer

See the table: deterministic methods (a)–(c) are fast/accurate but either lose uncertainty or do not scale; sampling methods (d)–(e) give the full posterior in high dimensions but need checking and/or tuning.

Takeaway & common mistakes

In practice they are combined: MAP gives starting values, the CLT gives proposal step sizes, and MCMC does the final job.

Exercise 3.2 — MAP and posterior mean with known variances

Exercise 2 (exact text from the book)

Assume that $Y_i|\mu \overset{indep}{\sim} \text{Normal}(\mu, \sigma_i^2)$ for $i \in \{1, \dots, n\}$, with σ_i known and improper prior distribution $\pi(\mu) = 1$ for all μ.

(a) Give a formula for the MAP estimator for μ.

(b) We observe $n = 3$, $Y_1 = 12$, $Y_2 = 10$, $Y_3 = 22$, $\sigma_1 = \sigma_2 = 3$ and $\sigma_3 = 10$; compute the MAP estimate of μ.

(c) Use numerical integration to compute the posterior mean of μ.

(d) Plot the posterior distribution of μ, and indicate the MAP and the posterior mean estimates on the plot.

In plain words

Three measurements of the same quantity μ have different accuracies (the third is much noisier). Find the most likely value of μ, then its posterior mean, and draw the posterior.

Background you need

The normal density is $\frac{1}{\sqrt{2\pi}\sigma}\exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\}$. Its log is $-\frac{(y-\mu)^2}{2\sigma^2} + (\text{terms without } \mu)$.

Key idea

With a flat prior, the posterior is proportional to the likelihood, so the MAP is the maximum-likelihood estimate — a **precision-weighted average** in which accurate measurements count more.

Solution

(a) Step 1 — write the log posterior.

With $\pi(\mu) = 1$, $\log \pi = 0$, so

$$\log p(\mu \mid \mathbf{Y}) = \sum_{i=1}^{n} \log f(Y_i \mid \mu) + \text{const} = -\sum_{i=1}^{n} \frac{(Y_i - \mu)^2}{2\sigma_i^2} + \text{const}.$$

(a) Step 2 — differentiate and set to zero.

$$\frac{\mathrm{d}}{\mathrm{d}\mu} \log p = \sum_{i=1}^{n} \frac{2(Y_i - \mu)}{2\sigma_i^2} = \sum_{i=1}^{n} \frac{Y_i - \mu}{\sigma_i^2} = 0.$$

Why? The derivative of $-(Y_i - \mu)^2$ with respect to μ is $+2(Y_i - \mu)$ (chain rule: the inner derivative of $Y_i - \mu$ is -1).

(a) Step 3 — solve for μ.

Write $w_i = 1/\sigma_i^2$ (the *precision* of measurement i). Then $\sum_i w_i Y_i - \mu \sum_i w_i = 0$, so

$$\hat{\mu}_{MAP} = \frac{\sum_{i=1}^{n} w_i Y_i}{\sum_{i=1}^{n} w_i}, \qquad w_i = \frac{1}{\sigma_i^2}.$$

(a) Step 4 — check it is a maximum.

$\frac{\mathrm{d}^2}{\mathrm{d}\mu^2}\log p = -\sum_i w_i < 0$, so the curve bends down: a maximum.

(b) Plug in the numbers.

$$w_1 = w_2 = \tfrac{1}{9} = 0.1111, \qquad w_3 = \tfrac{1}{100} = 0.01, \qquad \sum w_i = 0.2322,$$

$$\sum w_i Y_i = \tfrac{12}{9} + \tfrac{10}{9} + \tfrac{22}{100} = 1.3333 + 1.1111 + 0.2200 = 2.6644, \qquad \hat{\mu}_{MAP} = \frac{2.6644}{0.2322} = \mathbf{11.474}.$$

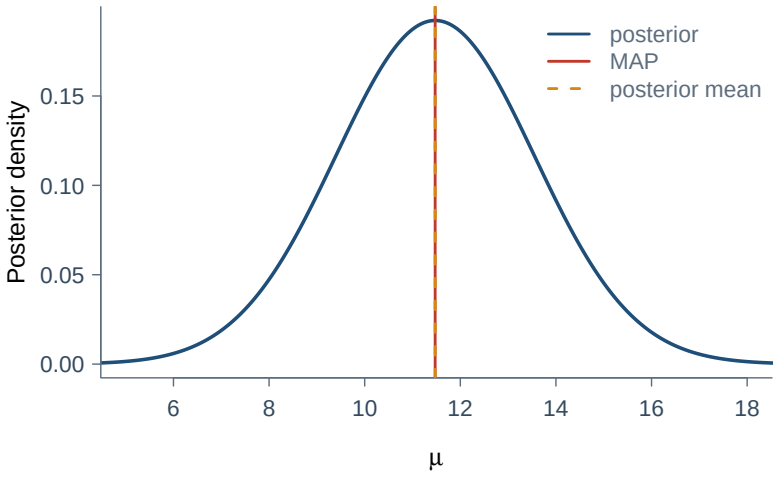
Why? The noisy third value (22) gets less than 5% of the total weight, so the estimate sits close to the average of 12 and 10.

(c) Numerical integration.

Evaluate the unnormalised posterior $g(\mu) = \exp\{\log p(\mu \mid \mathbf{Y})\}$ on a fine grid $\mu_1 < \cdots < \mu_G$ (5,001 points from 0 to 25, spacing Δ). Then

$$p(\mu_j \mid \mathbf{Y}) \approx \frac{g(\mu_j)}{\sum_k g(\mu_k)\Delta}, \qquad \mathrm{E}(\mu \mid \mathbf{Y}) \approx \sum_j \mu_j \, p(\mu_j \mid \mathbf{Y})\,\Delta = \mathbf{11.474}.$$

(d) Plot.



The two vertical lines lie exactly on top of each other.

Sanity check

The log posterior in Step 1 is a *quadratic* in μ, so the posterior is exactly normal: $\mu \mid \mathbf{Y} \sim \mathrm{Normal}\big(\hat{\mu}_{MAP},\ 1/\sum w_i\big) = \mathrm{Normal}(11.474,\ 2.075^2)$. For a symmetric distribution the mode (MAP) equals the mean, which is why (b) and (c) agree.

Answer

(a) $\hat{\mu}_{MAP} = \sum_i Y_i\sigma_i^{-2} \big/ \sum_i \sigma_i^{-2}$ (precision-weighted mean). (b) 11.474. (c) 11.474. (d) See the figure: the posterior is $\mathrm{Normal}(11.47,\ 2.08^2)$.

⚠ Takeaway & common mistakes

Never average measurements of different quality with equal weights: the simple mean $(12 + 10 + 22)/3 = 14.7$ is pulled far off by the least reliable value.

Exercise 3.3 — Poisson counts with exposures: MAP and the Bayesian CLT

Exercise 3 (exact text from the book)

Assume $Y_i|\lambda \overset{indep}{\sim} \text{Poisson}(N_i\lambda)$ for $i = 1, \ldots, n$.

(a) Identify a conjugate prior for λ, and derive the posterior that follows from this prior.

(b) Using the prior $\lambda \sim \text{Uniform}(0, 20)$, derive the MAP estimator of λ.

(c) Using the prior $\lambda \sim \text{Uniform}(0, 20)$, plot the posterior on a grid of λ assuming $n = 2$, $N_1 = 50$, $N_2 = 100$, $Y_1 = 12$, and $Y_2 = 25$, and show that the MAP estimate is indeed the maximizer.

(d) Use the Bayesian CLT to approximate the posterior of λ under the setting of (c), plot the approximate posterior, and compare the plot with the plot from (c).

In plain words

Unit i has size N_i (e.g. population or time) and produces Y_i events at rate λ per unit. Find a conjugate prior, the MAP under a uniform prior, and compare the exact posterior with its normal approximation.

Background you need

Poisson PMF: $\text{Prob}(Y = y) = \dfrac{e^{-m}m^y}{y!}$ with mean m. Gamma density: $\pi(\lambda) \propto \lambda^{a-1}e^{-b\lambda}$, mean a/b, variance a/b^2.

Key idea

As a function of λ, the likelihood is $\lambda^{\sum Y_i}e^{-\lambda\sum N_i}$ — a Gamma kernel. Only two numbers matter: total events $\sum Y_i$ and total exposure $\sum N_i$.

Solution

(a) Step 1 — the likelihood as a function of λ.

$$f(\mathbf{Y} \mid \lambda) = \prod_{i=1}^{n} \frac{e^{-N_i\lambda}(N_i\lambda)^{Y_i}}{Y_i!} = \underbrace{\prod_i \frac{N_i^{Y_i}}{Y_i!}}_{\text{no } \lambda} \lambda^{\sum Y_i} e^{-\lambda\sum N_i} \;\propto\; \lambda^{\sum Y_i}e^{-\lambda\sum N_i}.$$

(a) Step 2 — multiply by a Gamma prior.

$$p(\lambda \mid \mathbf{Y}) \propto \lambda^{\sum Y_i}e^{-\lambda\sum N_i} \cdot \lambda^{a-1}e^{-b\lambda} = \lambda^{(a+\sum Y_i)-1}e^{-(b+\sum N_i)\lambda}.$$

This is again a Gamma kernel, so the **Gamma prior is conjugate**: $\lambda \mid \mathbf{Y} \sim \text{Gamma}\big(a+\sum Y_i,\ b+\sum N_i\big)$.

(b) MAP under $\text{Uniform}(0, 20)$.

The prior is constant on $(0, 20)$, so on that interval $\log p = S_Y \log\lambda - S_N\lambda + \text{const}$ with $S_Y = \sum Y_i$, $S_N = \sum N_i$.

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\log p = \frac{S_Y}{\lambda} - S_N = 0 \;\Rightarrow\; \lambda = \frac{S_Y}{S_N}.$$

Second derivative $-S_Y/\lambda^2 < 0$: a maximum. Because the prior cuts off at 20,

$$\hat\lambda_{MAP} = \min\Big(\frac{\sum Y_i}{\sum N_i},\ 20\Big).$$

Why? If the unconstrained peak were above 20, the posterior would be increasing all the way up to the boundary, so its highest point would be at 20.

› (c) Numbers and grid plot.

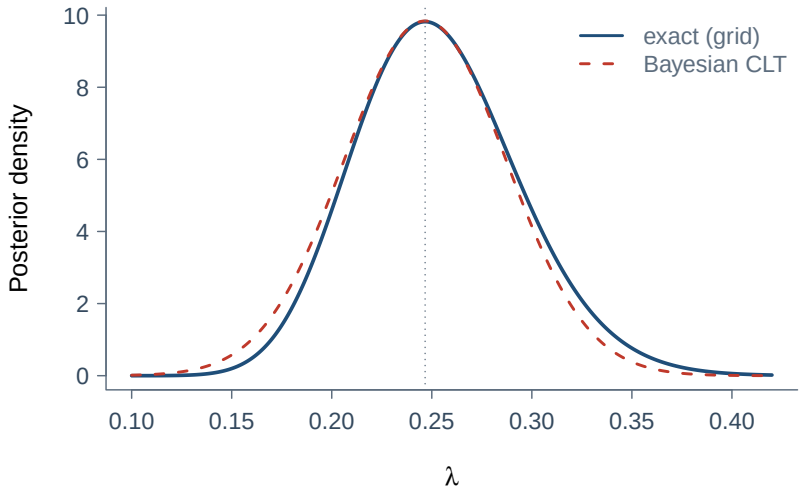
$S_Y = 12 + 25 = 37$, $S_N = 50 + 100 = 150$, so $\hat{\lambda}_{MAP} = 37/150 = \mathbf{0.2467}$. On a grid of 2,001 values the largest posterior value occurs at $\lambda = 0.2467$ — the same number (dotted line below).

› (d) Bayesian CLT.

Curvature at the MAP: $-\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} \log p = \frac{S_Y}{\lambda^2}$, evaluated at $\hat{\lambda}$:

$$\frac{S_Y}{\hat{\lambda}^2} = \frac{S_Y}{(S_Y/S_N)^2} = \frac{S_N^2}{S_Y} \quad \Rightarrow \quad \mathrm{Var} \approx \frac{S_Y}{S_N^2} = \frac{37}{150^2} = 0.001644, \quad \mathrm{SD} = 0.0406.$$

So $\lambda \mid \mathbf{Y} \approx \mathrm{Normal}(0.2467,\ 0.0406^2)$.



The exact posterior (a $\mathrm{Gamma}(38, 150)$ truncated at 20) is very slightly right-skewed; the normal approximation is symmetric but otherwise almost identical.

Sanity check

The exact posterior SD is $\sqrt{38}/150 = 0.0411$, very close to the CLT's 0.0406. With 37 events, the CLT is already accurate.

Answer

(a) $\lambda \sim \mathrm{Gamma}(a, b) \Rightarrow \lambda \mid \mathbf{Y} \sim \mathrm{Gamma}(a + \sum Y_i,\ b + \sum N_i)$. (b) $\hat{\lambda}_{MAP} = \min(\sum Y_i / \sum N_i,\ 20)$. (c) $\hat{\lambda}_{MAP} = 37/150 = 0.2467$, confirmed as the grid maximiser. (d) $\mathrm{Normal}(0.2467,\ 0.0406^2)$, nearly identical to the exact posterior.

Takeaway & common mistakes

The CLT works well here because there are many events (37). With, say, 2 events the Gamma posterior would be clearly skewed and the normal approximation could even put mass below zero.

Exercise 3.4 — A Gibbs sampler for a hierarchical variance model

Exercise 4 (exact text from the book)

Consider the model $Y_i|\sigma_i^2 \overset{indep}{\sim} \text{Normal}(0, \sigma_i^2)$ for $i = 1, \ldots, n$ where $\sigma_i^2|b \sim \text{InvGamma}(a, b)$ and $b \sim \text{Gamma}(1, 1)$.

(a) Derive the full conditional posterior distributions for σ_1^2 and b.

(b) Write pseudocode for Gibbs sampling; i.e., describe in detail each step of the Gibbs sampling algorithm.

(c) Write your own Gibbs sampling code (not in JAGS), and plot the marginal posterior density for each parameter. Assume $n = 10$, $a = 10$ and $Y_i = i$ for $i = 1, \ldots, 10$.

(d) Repeat the analysis with $a = 1$, and comment on the convergence of the MCMC chain.

(e) Implement this model in (c) using JAGS, and compare the results with the results in (c).

In plain words

Each observation has its own variance σ_i^2; the variances share a common scale b. Derive the Gibbs updates, code them, and compare two values of the prior shape a.

Background you need

$X \sim \text{InvGamma}(a, b)$ means $1/X \sim \text{Gamma}(a, b)$, with density $\pi(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} e^{-b/x}$ and mean $b/(a-1)$ for $a > 1$. Its kernel in x is $x^{-a-1}e^{-b/x}$; its kernel in b is $b^a e^{-b/x}$.

Key idea

To find a full conditional, write the joint density and keep only the factors containing the parameter of interest. Here both full conditionals turn out to be standard distributions, so plain Gibbs works.

Solution

(a) The joint posterior.

$$p(\sigma_1^2, \ldots, \sigma_n^2, b \mid \mathbf{Y}) \propto \prod_{i=1}^{n} \Big[\underbrace{(\sigma_i^2)^{-1/2} e^{-Y_i^2/(2\sigma_i^2)}}_{\text{likelihood}} \underbrace{\tfrac{b^a}{\Gamma(a)} (\sigma_i^2)^{-a-1} e^{-b/\sigma_i^2}}_{\text{prior of } \sigma_i^2} \Big] \underbrace{e^{-b}}_{\text{prior of } b}.$$

(a) Full conditional of σ_1^2.

Keep only factors with σ_1^2 (the $i = 1$ term):

$$p(\sigma_1^2 \mid \cdot) \propto (\sigma_1^2)^{-1/2} e^{-Y_1^2/(2\sigma_1^2)} (\sigma_1^2)^{-a-1} e^{-b/\sigma_1^2} = (\sigma_1^2)^{-(a+\frac{1}{2})-1} \exp\Big\{ -\frac{b + Y_1^2/2}{\sigma_1^2} \Big\}.$$

Why? Add the exponents of σ_1^2: $-\frac{1}{2} - a - 1 = -(a + \frac{1}{2}) - 1$; add the numerators in the exponential: $\frac{Y_1^2}{2} + b$.

This is the inverse-gamma kernel, so

$$\sigma_1^2 \mid \cdot \sim \text{InvGamma}\big(a + \tfrac{1}{2},\ b + \tfrac{Y_1^2}{2}\big) \qquad \text{(and the same for every } i\text{)}.$$

(a) Full conditional of b.

Keep factors with b: one $b^a e^{-b/\sigma_i^2}$ per observation, plus the prior e^{-b}:

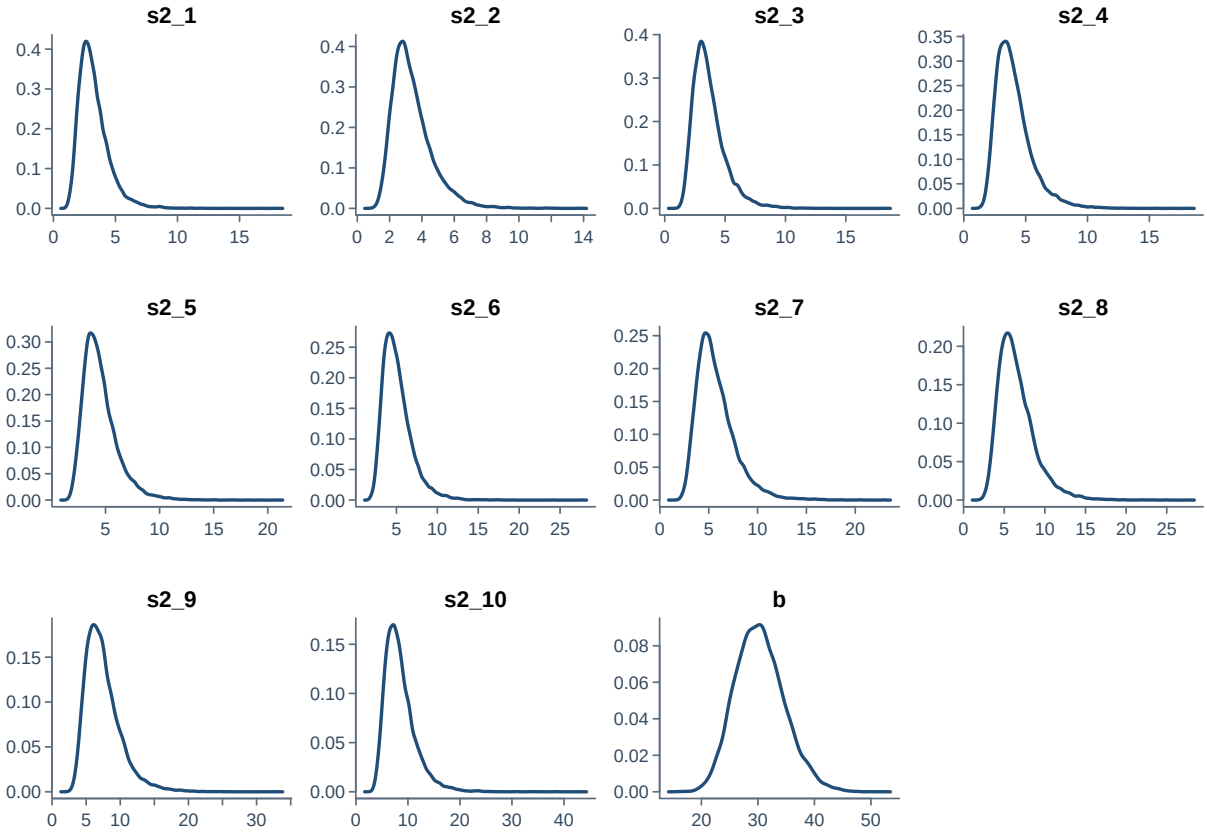
$$p(b \mid \cdot) \propto \prod_{i=1}^{n} b^a e^{-b/\sigma_i^2} \cdot e^{-b} = b^{na} \exp\Big\{ -b\Big(1 + \sum_i \frac{1}{\sigma_i^2}\Big)\Big\} \;\Rightarrow\; b \mid \cdot \sim \text{Gamma}\Big(na + 1,\; 1 + \sum_{i=1}^{n} \frac{1}{\sigma_i^2}\Big).$$

› (b) Pseudocode.

1. **Initialise**: $\sigma_i^{2(0)} = Y_i^2 + 1$ for all i, $b^{(0)} = 1$.
2. **For** $s = 1, \dots, S$:
   (i) for $i = 1, \dots, n$: draw $\sigma_i^{2(s)} \sim \text{InvGamma}\big(a + \tfrac{1}{2},\; b^{(s-1)} + Y_i^2/2\big)$ (in R: `1/rgamma(n, a + 0.5, b + Y2/2)`);
   (ii) draw $b^{(s)} \sim \text{Gamma}\big(na + 1,\; 1 + \sum_i 1/\sigma_i^{2(s)}\big)$;
   (iii) store $(\sigma_1^{2(s)}, \dots, \sigma_n^{2(s)}, b^{(s)})$.
3. **Discard** the first B draws (burn-in) and use the rest as posterior samples. Check trace plots and the effective sample size.

› (c) Results for $a = 10$

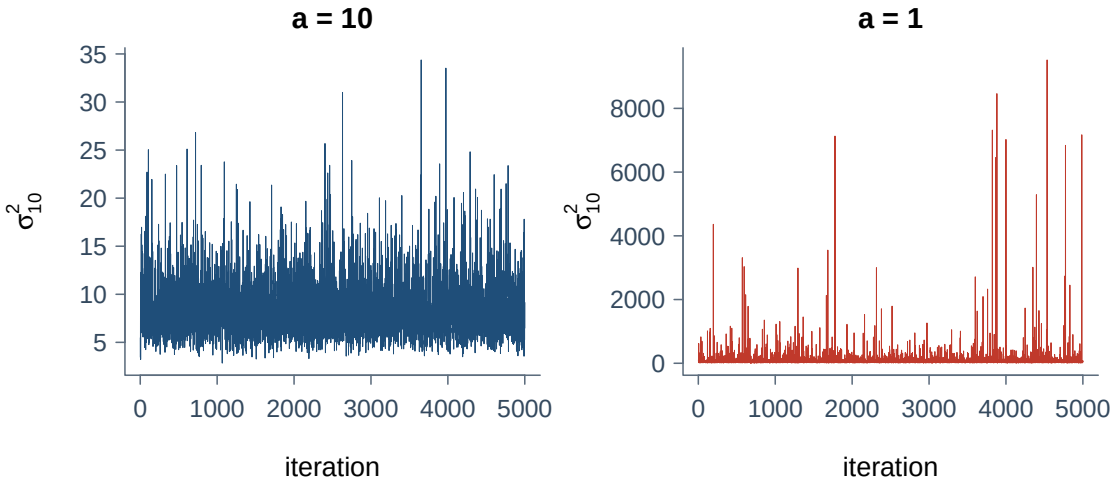
(20,000 iterations, first 2,000 discarded):



Posterior means: $b \approx 30.4$, $\sigma_1^2 \approx 3.25$, $\sigma_{10}^2 \approx 8.50$. With $a = 10$ the prior is fairly tight around $b/a \approx 3$, so even $Y_{10} = 10$ (which on its own suggests $\sigma_{10}^2 \approx 100$) is pulled down to about 8.5.

› (d) $a = 1$ and convergence.

Posterior means: $b \approx 5.6$, $\sigma_1^2 \approx 11.6$, $\sigma_{10}^2 \approx 112$. The chain itself mixes well (ESS for b: 10,451 out of 18,000), but the trace plot of σ_{10}^2 shows occasional huge spikes (note the very different vertical scales):



Why? With $a = 1$ the full conditional of σ_i^2 is $\text{InvGamma}(1.5, \cdot)$, which has a finite mean but an *infinite variance*. Rare enormous draws are a genuine feature of the posterior, not a convergence failure — but they make Monte Carlo averages of σ_i^2 unstable, so we need many more draws (or should report medians).

(e) JAGS.

The same model in JAGS (prior on the precision $1/\sigma_i^2 \sim \text{Gamma}(a, b)$, which is equivalent):

```
model{
  for(i in 1:n){
    Y[i] ~ dnorm(0, prec[i])      # JAGS uses precision = 1/variance
    prec[i] ~ dgamma(a, b)        # => sigma2[i] ~ InvGamma(a, b)
    sigma2[i] <- 1 / prec[i]
  }
  b ~ dgamma(1, 1)
}
```

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	$a = 10$			$a = 1$		
	JAGS mean	own Gibbs	$\hat{R}$	JAGS mean	own Gibbs	$\hat{R}$
b	30.26	30.35	1.00	5.55	5.56	1.00
σ_1^2	3.24	3.25	1.00	12.1	11.6	1.07
σ_{10}^2	8.46	8.50	1.00	107	112	1.02

The results agree. For $a = 1$ even JAGS shows $\hat{R}$ slightly above 1 and posterior SDs of 82 and 563 for $\sigma_1^2, \sigma_{10}^2$: the heavy tails make means of σ_i^2 hard to estimate, exactly as discussed in (d).

Answer

(a) $\sigma_i^2 \mid \cdot \sim \text{InvGamma}(a + \frac{1}{2},\ b + \frac{Y_i^2}{2})$ and $b \mid \cdot \sim \text{Gamma}(na + 1,\ 1 + \sum_i \sigma_i^{-2})$. (b) Alternate the two draws (pseudocode above). (c) $a = 10$: strong pooling, $\text{E}(b \mid \mathbf{Y}) \approx 30$. (d) $a = 1$: weak pooling; good mixing but heavy-tailed draws of σ_i^2. (e) JAGS agrees.

Takeaway & common mistakes

"Good mixing" (high ESS) and "stable averages" are different things: if the posterior has very heavy tails, even independent draws give noisy means.

Exercise 3.5 — Gibbs sampling for a two-group comparison

Exercise 5 (exact text from the book)

Consider the model $Y_i|\mu,\sigma^2 \sim \text{Normal}(\mu,\sigma^2)$ for $i=1,\ldots,n$ and $Y_i|\mu,\delta,\sigma^2 \sim \text{Normal}(\mu+\delta,\sigma^2)$ for $i=n+1,\ldots,n+m$, where $\mu,\delta \sim \text{Normal}(0,100^2)$ and $\sigma^2 \sim \text{InvGamma}(0.01,0.01)$.

(a) Give an example of a real experiment for which this would be an appropriate model.

(b) Derive the full conditional posterior distributions for μ, δ and σ^2.

(c) Simulate a dataset from this model with $n=m=50$, $\mu=10$, $\delta=1$ and $\sigma=2$. Write your own Gibbs sampling code (not in JAGS) to fit the model above to the simulated data, and plot the marginal posterior density for each parameter. Are you able to recover the true values reasonably well?

(d) Implement this model using JAGS, and compare the results with the results in (c).

In plain words

A control group has mean μ; a treatment group has mean $\mu+\delta$; both have the same noise level σ. Derive the Gibbs updates and check them on simulated data.

Background you need

Completing the square. If a log density in x has the form $-\frac{A}{2}x^2 + Bx + \text{const}$, then $x \sim \text{Normal}(B/A,\ 1/A)$. So for normal full conditionals we only need to collect the coefficient of x^2 (the precision A) and of x (the term B).

Key idea

Write $g_i = 1$ for treated units and $g_i = 0$ for controls, so $Y_i \sim \text{Normal}(\mu+\delta g_i,\sigma^2)$ for every i. Each full conditional is normal or inverse-gamma.

Solution

(a) Example.

A clinical trial of a blood-pressure drug: n patients get a placebo (mean μ), m get the drug (mean $\mu+\delta$); δ is the treatment effect, and the spread σ is assumed equal in both groups. (Also: crop yield without/with a fertiliser.)

(b) Full conditional of μ.

Let $N=n+m$. The terms with μ are

$$-\frac{1}{2\sigma^2}\sum_{i=1}^{N}(Y_i-\delta g_i-\mu)^2 - \frac{\mu^2}{2\cdot 100^2}.$$

Coefficient of μ^2: $A=\frac{N}{\sigma^2}+\frac{1}{100^2}$. Coefficient of μ: $B=\frac{1}{\sigma^2}\sum_i(Y_i-\delta g_i)$. Hence

$$\mu \mid \cdot \sim \text{Normal}\Big(\frac{\sum_i(Y_i-\delta g_i)/\sigma^2}{N/\sigma^2+1/100^2},\ \frac{1}{N/\sigma^2+1/100^2}\Big).$$

Why? Expanding $(Y_i-\delta g_i-\mu)^2=\mu^2-2\mu(Y_i-\delta g_i)+\ldots$; summing over i and multiplying by $-\frac{1}{2\sigma^2}$ gives $-\frac{N}{2\sigma^2}\mu^2+\frac{\sum(Y_i-\delta g_i)}{\sigma^2}\mu$.

(b) Full conditional of δ.

Only the treated observations contain δ:

$$\delta \mid \cdot \sim \text{Normal}\Big(\frac{\sum_{i>n}(Y_i-\mu)/\sigma^2}{m/\sigma^2+1/100^2},\ \frac{1}{m/\sigma^2+1/100^2}\Big).$$

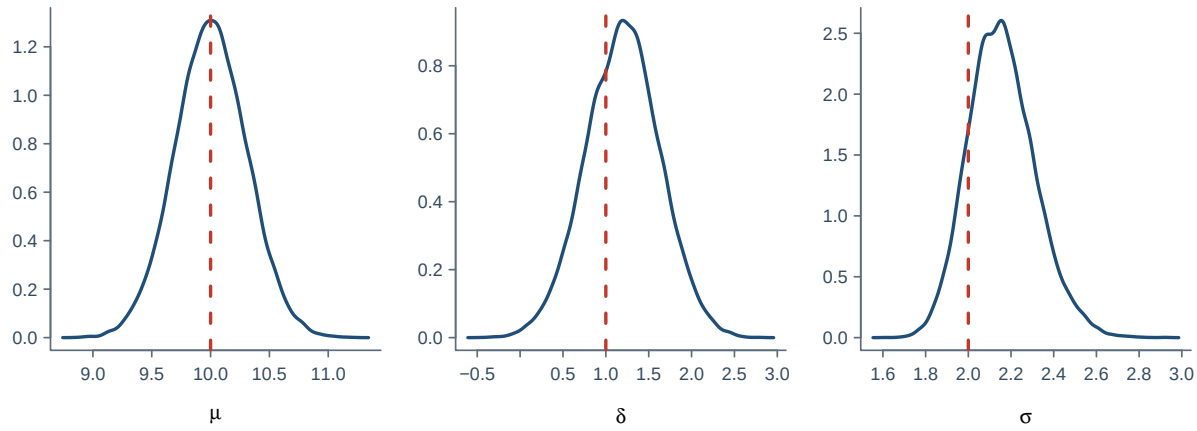
› (b) Full conditional of σ^2.

$$p(\sigma^2 \mid \cdot) \propto (\sigma^2)^{-N/2}\exp\Big\{-\frac{\text{SSE}}{2\sigma^2}\Big\}(\sigma^2)^{-0.01-1}e^{-0.01/\sigma^2}, \qquad \text{SSE}=\sum_i (Y_i-\mu-\delta g_i)^2,$$

so $\sigma^2 \mid \cdot \sim \text{InvGamma}\big(0.01+\frac{N}{2},\ 0.01+\frac{\text{SSE}}{2}\big)$.

› (c) Simulation and Gibbs sampler.

We simulated 50 values from $\text{N}(10, 2^2)$ and 50 from $\text{N}(11, 2^2)$, then ran 20,000 Gibbs iterations (1,000 burn-in), cycling through the three updates above.



Parameter	Truth	Posterior mean	SD	95% interval
μ	10	10.005	0.305	(9.41, 10.61)
δ	1	1.199	0.431	(0.35, 2.04)
σ	2	2.157	0.156	(1.88, 2.49)

All three true values (red dashed lines) fall well inside the 95% intervals.

Sanity check

With such vague priors the posterior means should match the classical estimates: the control-group mean is 10.010, the difference in group means is 1.190 and the pooled SD is 2.141 — essentially the same as the table.

› (d) JAGS.

```
model{
  for(i in 1:N){ Y[i] ~ dnorm(mu + delta * group[i], tau) }
  mu    ~ dnorm(0, 1 / 100^2)
  delta ~ dnorm(0, 1 / 100^2)
  tau   ~ dgamma(0.01, 0.01)        # tau = 1 / sigma^2
  sigma <- 1 / sqrt(tau)
}
```

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	JAGS mean	JAGS 95% interval	$\hat{R}$	own Gibbs mean
μ	10.010	(9.40, 10.62)	1.00	10.005
δ	1.191	(0.34, 2.04)	1.00	1.199
σ	2.158	(1.88, 2.49)	1.00	2.157

Identical up to Monte Carlo error.

Answer

(a) Placebo vs. treatment trial. (b) $\mu \mid \cdot$ and $\delta \mid \cdot$ normal (formulas above), $\sigma^2 \mid \cdot \sim \text{InvGamma}(0.01 + \frac{N}{2},\ 0.01 + \frac{\text{SSE}}{2})$. (c) Yes — all true values are recovered within the 95% intervals. (d) JAGS gives the same answers.

Takeaway & common mistakes

The treatment effect δ has the widest interval (it is a difference of two noisy means), so 50 per group is only just enough to see that $\delta > 0$.

Exercise 3.6 — Metropolis-within-Gibbs

Exercise 6 (exact text from the book)

Consider the model $Y|\theta, b \sim \text{Binomial}(n, \theta)$, $\theta|b \sim \text{Beta}\{\exp(b), \exp(b)\}$ and $b \sim \text{Normal}(0, 1)$.

(a) Specify initial values for θ and b.

(b) What is the full conditional distribution of θ?

(c) The full conditional distribution of b does not have a nice form and thus cannot be updated using Gibbs sampling. Sketch a Metropolis-within-Gibbs sampler for the posterior of (θ, b).

(d) Implement your Metropolis-within-Gibbs sampler with $n = 100$ and $Y = 10$, and summarize the posterior.

In plain words

The prior on the success probability is a symmetric Beta whose concentration e^b is itself unknown. Update θ by Gibbs and b by Metropolis.

Background you need

$\text{Beta}(c, c)$ is symmetric around 0.5; larger c means more concentrated near 0.5. Its density is $\frac{\Gamma(2c)}{\Gamma(c)^2}\theta^{c-1}(1-\theta)^{c-1}$ — note the normalising constant depends on c, so it must be kept when c is unknown.

Key idea

Mix the two samplers: draw a parameter exactly when its full conditional is standard (Gibbs), and use a Metropolis step for the one that is not.

Solution

(a) Initial values.

$\theta^{(0)} = Y/n = 0.1$ (the sample proportion) and $b^{(0)} = 0$ (the prior mean, i.e. a $\text{Beta}(1, 1)$ uniform prior).

(b) Full conditional of θ.

$$p(\theta \mid b, Y) \propto \theta^{Y}(1-\theta)^{n-Y} \cdot \theta^{e^b-1}(1-\theta)^{e^b-1} \;\Rightarrow\; \theta \mid b, Y \sim \text{Beta}\big(Y + e^b,\; n - Y + e^b\big).$$

(c) Full conditional of b and the sampler.

Keep everything that involves b — including the Beta normalising constant:

$$\log p(b \mid \theta, Y) = -\frac{b^2}{2} + \log\Gamma(2e^b) - 2\log\Gamma(e^b) + (e^b - 1)\log[\theta(1-\theta)] + \text{const.}$$

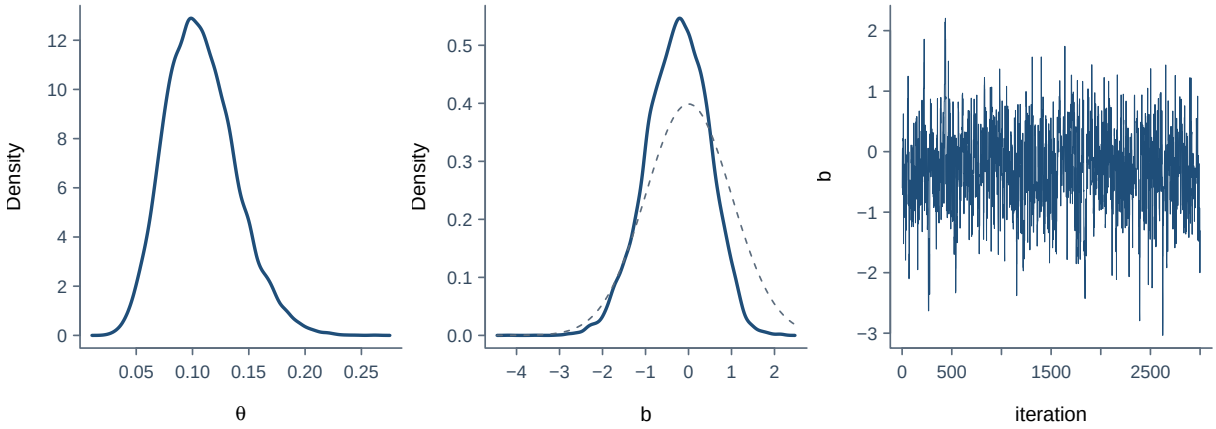
This is not a standard density, so we use a random-walk Metropolis step. One iteration:

1. Gibbs: draw $\theta \sim \text{Beta}(Y + e^b,\; n - Y + e^b)$.
2. Metropolis: propose $b^* \sim \text{Normal}(b,\; s^2)$; compute $\log R = \log p(b^* \mid \theta, Y) - \log p(b \mid \theta, Y)$; draw $U \sim \text{Uniform}(0, 1)$; if $\log U < \log R$ set $b = b^*$, otherwise keep b.

(d) Results

(step $s = 1$, 20,000 iterations, acceptance rate 61%):

	Mean	SD	95% interval	ESS
θ	0.108	0.031	(0.054, 0.175)	16,791
b	-0.26	0.72	$(-1.74,\ 1.06)$	3,213



Sanity check

The posterior of θ is close to the $\mathrm{Beta}(11, 91)$ we would get with a uniform prior (mean 0.108) — the data dominate. The posterior of b (middle panel, solid) is only slightly shifted from its $\mathrm{N}(0, 1)$ prior (dashed): a single binomial count says very little about the prior's concentration.

Answer

(a) $\theta^{(0)} = 0.1$, $b^{(0)} = 0$. (b) $\theta \mid b, Y \sim \mathrm{Beta}(Y + e^b,\ n - Y + e^b)$. (c) Gibbs for θ, random-walk Metropolis for b. (d) θ: mean 0.108, 95% CI (0.054, 0.175); b: mean -0.26, close to its prior.

Takeaway & common mistakes

Do not drop normalising constants that depend on the parameter you are updating — here $\Gamma(2e^b)/\Gamma(e^b)^2$ is essential in the Metropolis ratio for b.

Exercise 3.7 — Shrinking clutch shooting toward each player's overall rate

Exercise 7 (exact text from the book)

Fit the following model to the NBA free throw data in the table in Exercise 22 in Chapter 1:

$$Y_i|\theta_i \sim \text{Binomial}(n_i, \theta_i) \quad \text{and} \quad \theta_i|m \sim \text{Beta}[\exp(m)q_i, \exp(m)(1-q_i)],$$

where Y_i is the number of made clutch shots for player $i = 1, \ldots, 10$, n_i is the number of attempted clutch shots, $q_i \in (0, 1)$ is the overall proportion, and $m \sim \text{Normal}(0, 10)$.

(a) Explain why this is a reasonable prior for θ_i.

(b) Explain the role of m in the prior.

(c) Derive the full conditional posterior for θ_1.

(d) Write your own MCMC algorithm to compute a table of posterior means and 95% credible intervals for all 11 model parameters $(\theta_1, \ldots, \theta_{10}, m)$. Turn in commented code.

(e) Fit the same model in JAGS. Turn in commented code, and comment on whether the two algorithms returned the same results.

(f) What are the advantages and disadvantages of writing your own code as opposed to using JAGS in this problem and in general?

In plain words

Instead of analysing each player alone (Chapter 1), centre each player's clutch prior on their overall free-throw percentage, and let the data decide how tightly to hold them there.

Background you need

For $\theta \sim \text{Beta}(\alpha, \beta)$: $\text{E}\theta = \frac{\alpha}{\alpha+\beta}$ and $\text{Var}\theta = \frac{\text{E}\theta(1-\text{E}\theta)}{\alpha+\beta+1}$. The sum $\alpha + \beta$ acts like a "prior sample size".

Key idea

Here $\alpha + \beta = e^m$, so m is the log prior sample size: it controls how much a player's clutch rate is allowed to differ from their overall rate.

Solution

(a) Why the prior is reasonable.

$$\text{E}(\theta_i \mid m) = \frac{e^m q_i}{e^m q_i + e^m(1-q_i)} = q_i.$$

The prior is centred on the player's overall percentage — the natural guess for their clutch percentage — and stays inside $(0, 1)$.

(b) The role of m.

$\text{Var}(\theta_i \mid m) = \dfrac{q_i(1-q_i)}{e^m + 1}$. Large m: tiny variance, so $\theta_i \approx q_i$ ("clutch is the same as overall"). Small m: wide prior, so each player's clutch rate is estimated from their clutch data alone. Because m is unknown, *the data decide* how much to shrink toward q_i.

(c) Full conditional of θ_1.

$$p(\theta_1 \mid \cdot) \propto \theta_1^{Y_1}(1-\theta_1)^{n_1-Y_1}\,\theta_1^{e^m q_1-1}(1-\theta_1)^{e^m(1-q_1)-1} \;\Rightarrow\; \theta_1 \mid \cdot \sim \text{Beta}\big(Y_1+e^m q_1,\; n_1-Y_1+e^m(1-q_1)\big).$$

› (d) The MCMC algorithm

(Gibbs for every θ_i, Metropolis for m), from `ch03.R`:

```
log_post_m ← function(m, theta, q) {          # log full conditional of m
  e ← exp(m)
  dnorm(m, 0, sqrt(10), log = TRUE) +         # prior N(0, variance 10)
    sum(dbeta(theta, e * q, e * (1 - q), log = TRUE))
}
for (it in seq_len(iters)) {
  e     ← exp(m)
  theta ← rbeta(10, y + e * q, n - y + e * (1 - q))     # Gibbs: all thetas at once
  cand  ← rnorm(1, m, 0.6)                              # Metropolis proposal for m
  if (log(runif(1)) < log_post_m(cand, theta, q) - log_post_m(m, theta, q)) m ← cand
  draws[it, ] ← c(theta, m)
}
```

30,000 iterations (2,000 burn-in), acceptance rate for m: 64%.

Parameter	Posterior mean	95% interval	ESS	Ch. 1 (no pooling) mean
θ Westbrook	0.847	(0.799, 0.892)	25,331	0.844
θ Harden	0.819	(0.745, 0.859)	953	0.753
θ Leonard	0.878	(0.830, 0.918)	25,385	0.862
θ James	0.678	(0.610, 0.749)	24,122	0.683
θ Thomas	0.908	(0.868, 0.940)	28,000	0.894
θ Curry	0.902	(0.855, 0.947)	18,438	0.893
θ Antetokounmpo	0.753	(0.670, 0.805)	2,447	0.674
θ Wall	0.802	(0.751, 0.851)	28,000	0.798
θ Davis	0.788	(0.715, 0.836)	3,928	0.732
θ Durant	0.869	(0.801, 0.915)	8,861	0.778
m	5.66	(3.35, 9.28)	264	—

The posterior of m is large ($e^{5.66} \approx 290$ pseudo-shots), so every clutch rate is pulled strongly toward the player's overall rate. Harden moves from 0.753 to 0.819, Durant (only 16 attempts) from 0.778 to 0.869.

› (e) JAGS.

```
model{
  for(i in 1:10){
    Y[i] ~ dbin(theta[i], n[i])
    theta[i] ~ dbeta(exp(m) * q[i], exp(m) * (1 - q[i]))
  }
  m ~ dnorm(0, 0.1)          # JAGS: precision 0.1 = variance 10
}
```

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	JAGS	own		JAGS	own
Westbrook	0.847	0.847	Curry	0.902	0.902
Harden	0.818	0.819	Antetokounmpo	0.753	0.753
Leonard	0.878	0.878	Wall	0.802	0.802
James	0.677	0.678	Davis	0.788	0.788
Thomas	0.907	0.908	Durant	0.869	0.869

m: JAGS 5.60 (3.39, 9.02), own 5.66 (3.35, 9.28); all $\hat{R} \leq 1.002$

The two algorithms return the same posterior means to three decimals.

(f) Own code vs. JAGS.

- **Own code** — advantages: complete control (e.g. updating all θ_i in one vectorised line), often faster, and you understand every step. Disadvantages: time-consuming, easy to make algebra or coding mistakes, must tune Metropolis steps yourself.
- **JAGS** — advantages: you only write the model; samplers and tuning are automatic, so changing the model takes seconds. Disadvantages: a black box, can be slow or mix badly for awkward models, and limited to the distributions JAGS knows.

Answer

(a) The prior mean of θ_i is the overall rate q_i. (b) e^m is a prior sample size controlling shrinkage toward q_i. (c) $\theta_1 \mid \cdot \sim \text{Beta}(Y_1 + e^m q_1,\ n_1 - Y_1 + e^m(1 - q_1))$. (d)–(e) Table above; $m \approx 5.7$ means strong shrinkage; JAGS agrees. (f) Control/speed vs. convenience/safety.

Takeaway & common mistakes

The data say clutch and overall shooting are very similar for most players, which is why m is large. The "clutch-choker" effect we saw for Harden in Chapter 1 shrinks substantially once we borrow strength across players.

Exercise 3.8 — Metropolis–Hastings for a gamma shape parameter

Exercise 8 (exact text from the book)

Assume the model $Y|\theta \sim \text{Gamma}(\theta, 1)$ and prior $\theta \sim \text{Uniform}(0, 10)$. Say we observe $Y = 5$. This is not a conjugate prior, and so you will use Metropolis-Hastings sampling to approximate the posterior of θ.

(a) Plot the posterior of θ for reference, i.e. (in R),

```
> theta <- seq(0,10,0.01)
> plot(theta,dgamma(5,theta,1),type="l")
```

(b) What is a reasonable candidate distribution for θ?

(c) Give a formula for the acceptance probability (preferably in R code). How will you handle candidates outside $(0, 10)$?

(d) How will you tune the candidate distribution? Be specific.

(e) Generate and plot samples from the posterior distribution, and compare results with the curve from (a)

In plain words

Build, tune and check a Metropolis sampler for a one-parameter posterior that we can also draw exactly, so we can see whether the sampler works.

Background you need

With a uniform prior on $(0, 10)$, the posterior is $p(\theta \mid Y) \propto \frac{Y^{\theta-1}e^{-Y}}{\Gamma(\theta)}$ for $0 < \theta < 10$ and 0 outside — exactly the curve `dgamma(5, theta, 1)` in part (a).

Key idea

A symmetric random-walk proposal makes the acceptance ratio the plain ratio of posterior densities. Proposals outside the prior support have posterior density 0, so they are automatically rejected.

Solution

(a) Reference curve.

The red curve in the figure of part (e) is `dgamma(5, theta, 1)` rescaled to integrate to 1.

(b) Candidate distribution.

A normal random walk: $\theta^* \sim \text{Normal}(\theta^{(s-1)}, s^2)$. It is symmetric ($q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$), so the proposal densities cancel.

(c) Acceptance probability.

$$\alpha = \min\left\{1, \frac{p(\theta^* \mid Y)}{p(\theta \mid Y)}\right\} = \min\left\{1, \frac{\texttt{dgamma}(5, \theta^*, 1)\,\mathbb{1}(0 < \theta^* < 10)}{\texttt{dgamma}(5, \theta, 1)}\right\}.$$

In R, on the log scale:

```
log_post <- function(theta) ifelse(theta > 0 & theta < 10, dgamma(5, theta, 1, log = TRUE),
    -Inf)
accept   <- log(runif(1)) < log_post(cand) - log_post(theta)
```

Why? A candidate outside $(0, 10)$ gets $\log p = -\infty$, so $\log R = -\infty$ and it is never accepted; the chain simply stays where it is for that iteration. Do *not* just discard such candidates and redraw — that would change the target distribution.

› (d) Tuning.

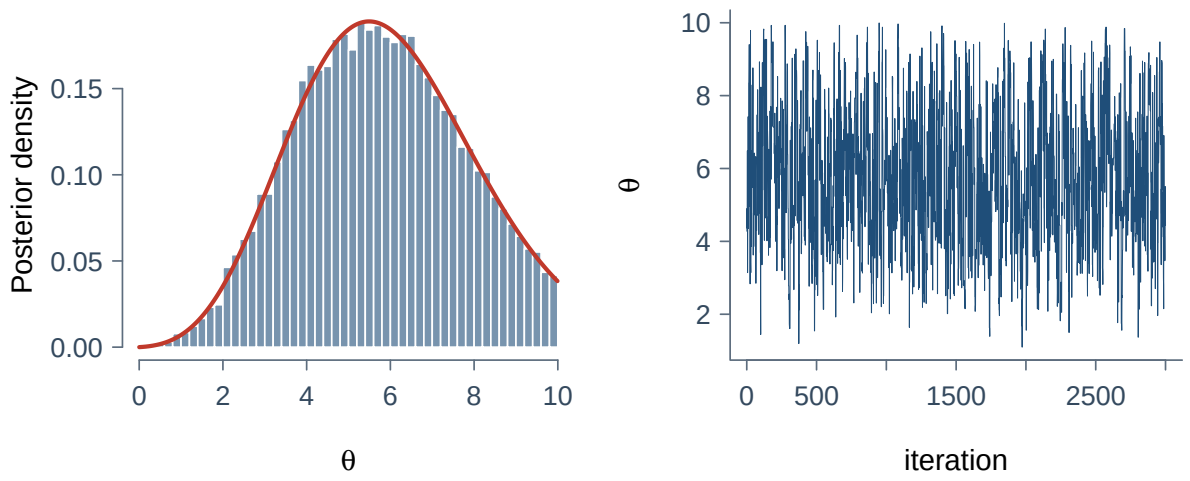
Try several step sizes and look at the acceptance rate and ESS (10,000 iterations each):

Step s	Acceptance	ESS	Verdict
0.2	0.97	26	steps too small: chain crawls
1.0	0.86	497	still small
2.5	**0.65**	**1,725**	**best**
8.0	0.30	1,628	many rejections

Rule: increase s if the acceptance rate is above ∼50%, decrease it if below ∼25%, and pick the value with the largest ESS. We use $s = 2.5$.

› (e) Samples.

50,000 iterations (1,000 burn-in):



The histogram matches the exact curve; the trace plot looks like a hairy caterpillar.

Sanity check

MCMC posterior mean 5.725 vs. 5.737 from the exact curve (numerical integration); 95% interval $(2.12,\ 9.46)$.

Answer

(b) $\theta^* \sim \mathrm{N}(\theta, s^2)$. (c) $\alpha = \min\{1, \texttt{dgamma(5,cand,1)/dgamma(5,theta,1)}\}$, and candidates outside $(0, 10)$ are rejected (chain stays put). (d) Tune s by acceptance rate/ESS; $s = 2.5$ works best. (e) Samples match the exact posterior (mean ≈ 5.7).

Takeaway & common mistakes

A very high acceptance rate is *not* good: it usually means the steps are tiny and the chain barely moves (ESS 26 at 97% acceptance!).

Exercise 3.9 — A Student-t model for the galaxies data

Exercise 9 (exact text from the book)

Open and plot the galaxies data in R using the code below,

```
> library(MASS)
> data(galaxies)
> ?galaxies
> Y <- galaxies
> n <- length(Y)
> hist(Y,breaks=25)
```

Model the observations $Y_1, \ldots, Y_{82}$ using the Student's t-distribution with location μ, scale σ and degrees of freedom k. Assume prior distributions $\mu \sim \text{Normal}(0, 10000^2)$, $1/\sigma^2 = \tau \sim \text{Gamma}(0.01, 0.01)$ and $k \sim \text{Uniform}(1, 30)$.

(a) Give reasonable initial values for each of the three parameters.

(b) Fit the model using JAGS. Report trace plots of each parameter, and discuss convergence.

(c) Graphically compare the t distribution with parameters set to their posterior mean with the observed data. Does the model fit the data well?

In plain words

Fit a heavy-tailed bell curve to the velocities of 82 galaxies, check that the sampler converged, and see whether a single bell curve describes the data.

Background you need

The t distribution as a scale mixture. If $\lambda_i \sim \text{Gamma}(k/2, k/2)$ and $Y_i \mid \lambda_i \sim \text{Normal}(\mu, \sigma^2/\lambda_i)$, then marginally $Y_i \sim t_k(\mu, \sigma)$. Small λ_i = an observation allowed to be far out. This makes almost every full conditional standard.

Key idea

Use the data to pick initial values (centre and spread). With the latent λ_i, the full conditionals are: μ normal, τ gamma, λ_i gamma; k needs a Metropolis step.

Solution

(a) Initial values.

$\mu^{(0)}$ = median of the data = 20,834; $\tau^{(0)} = 1/\text{var}(Y)$ (so $\sigma^{(0)} \approx 4{,}600$); $k^{(0)} = 5$ (moderately heavy tails). For a convergence check we also start chains far away: $(\mu, \sigma, k) = (15{,}000,\ 5{,}000,\ 25)$ and $(25{,}000,\ 1{,}000,\ 3)$.

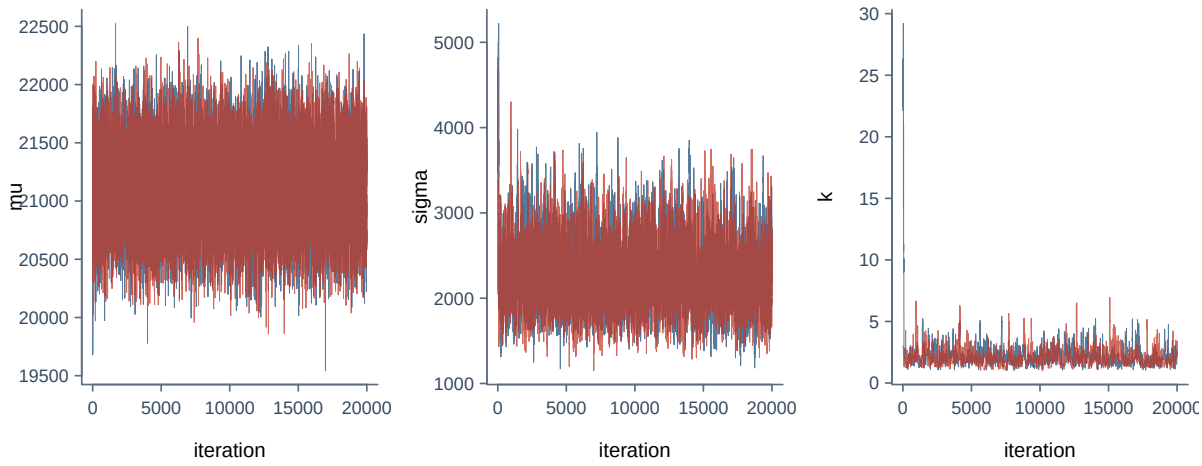
(b) Full conditionals used by our sampler.

$$\mu \mid \cdot \sim \text{N}\Big(\frac{\tau \sum \lambda_i Y_i}{\tau \sum \lambda_i + 10^{-8}},\ \frac{1}{\tau \sum \lambda_i + 10^{-8}}\Big), \quad \tau \mid \cdot \sim \text{Gamma}\Big(0.01 + \frac{n}{2},\ 0.01 + \frac{\sum \lambda_i (Y_i - \mu)^2}{2}\Big),$$

$$\lambda_i \mid \cdot \sim \text{Gamma}\Big(\frac{k+1}{2},\ \frac{k + \tau(Y_i - \mu)^2}{2}\Big), \qquad k: \text{ Metropolis step on } (1, 30).$$

(b) Trace plots

(two chains from the extreme starting values, 20,000 iterations each):



Both chains forget their starting values within a few hundred iterations and then overlap. $\hat{R} = 1.000$ for all three parameters; ESS: μ 13,284, σ 1,131, k 494 (after 2,000 burn-in per chain). k mixes slowest, as is typical for degrees of freedom, but all ESS are adequate.

	Mean	SD	95% interval
μ	21,115	325	(20,481, 21,757)
σ	2,240	349	(1,641, 3,008)
k	2.1	0.6	(1.2, 3.6)

JAGS (`Y[i] ~ dt(mu, tau, k)`):

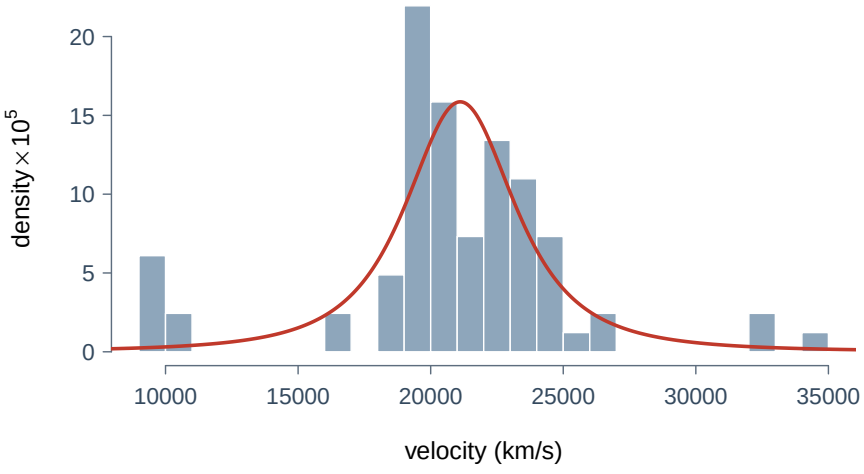
Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	Mean	95% interval	$\hat{R}$
μ	21,120	(20,490, 21,760)	1.000
σ	2,268	(1,651, 3,057)	1.000
k	2.16	(1.25, 3.82)	1.004

Same answer as our sampler.

› (c) Fit.



The fit is poor. The histogram has several clusters (around 10,000, 20,000–23,000 and 33,000 km/s); a single symmetric t curve cannot reproduce multiple peaks. The model "explains" the outer clusters by choosing very heavy tails ($k \approx 2$).

Answer

(a) $\mu^{(0)} = \text{median}(Y)$, $\tau^{(0)} = 1/\text{var}(Y)$, $k^{(0)} = 5$. (b) Converged ($\hat{R} = 1.00$, adequate ESS); $\mu \approx 21{,}100$, $\sigma \approx 2{,}240$, $k \approx 2.1$. (c) Poor fit: the data are multi-modal and a single t density cannot capture the clusters — a mixture model is needed.

Takeaway & common mistakes

Convergence of the sampler tells you nothing about whether the *model* is right. Always compare the fitted model with the data.

Exercise 3.10 — A Laplace model for the galaxies data, on a grid

Exercise 10 (exact text from the book)

Download the galaxy data

```
> library(MASS)
> data(galaxies)
> Y ← galaxies
```

Suppose $Y_i|\theta \overset{\text{iid}}{\sim} \text{Laplace}(\mu, \sigma)$ for $i = 1, \ldots, n$ where $\theta = (\mu, \sigma)$.

(a) Assuming the improper prior $\sigma \sim \text{Uniform}(0, 100000)$ and $\pi(\mu) = 1$ for all $\mu \in (-\infty, \infty)$, plot the joint posterior distribution of θ and the marginal posterior distributions of μ and σ.

(b) Compute the posterior mean of θ from your analysis in (a), and plot the Laplace PDF with these values against the observed data. Does the model fit the data well?

(c) Plot the posterior predictive distribution (PPD) for a new observation $Y^*|\theta \sim \text{Laplace}(\mu, \sigma)$. How do the mean and variance of the PPD compare to the mean and variance of the "plug-in" distribution from (b)?

In plain words

Fit a double-exponential (Laplace) curve with two unknowns by brute force on a 2-D grid, check the fit, and compare predicting with uncertainty vs. plugging in estimates.

Background you need

$\text{Laplace}(\mu, \sigma)$: density $\frac{1}{2\sigma}\exp\{-|y-\mu|/\sigma\}$, mean μ, variance $2\sigma^2$. A Laplace draw is $\mu + \sigma(E_1 - E_2)$ with $E_1, E_2 \overset{\text{iid}}{\sim} \text{Exponential}(1)$.

Key idea

With two parameters, **grid approximation** is easy and exact enough: evaluate the log posterior on a 300×300 grid of (μ, σ), exponentiate, normalise, and sum over rows/columns for the marginals.

Solution

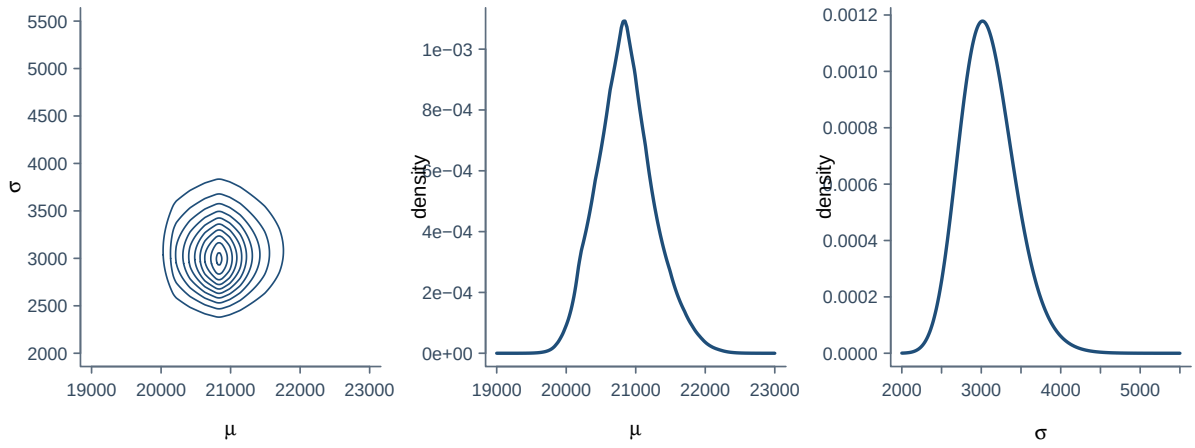
(a) Step 1 — the log posterior.

With flat priors (the upper bound 100,000 is far from where the likelihood lives),

$$\log p(\mu, \sigma \mid \mathbf{Y}) = -n \log \sigma - \frac{1}{\sigma} \sum_{i=1}^{n} |Y_i - \mu| + \text{const.}$$

(a) Step 2 — evaluate on a grid.

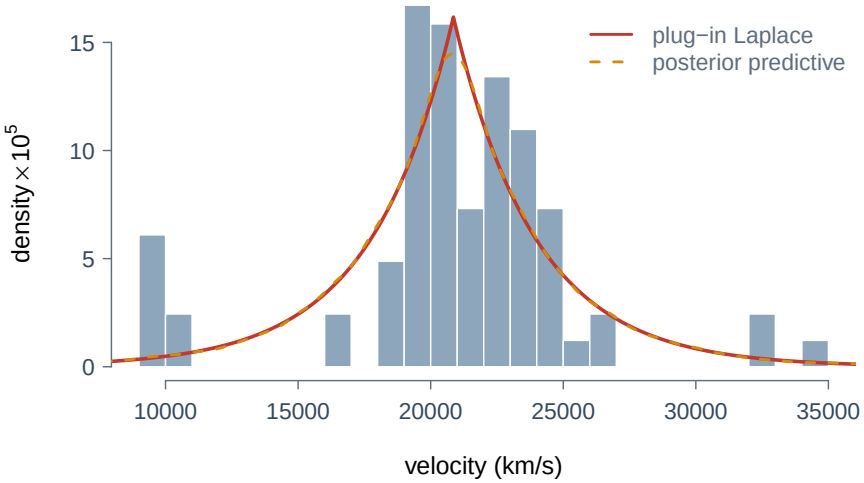
$\mu \in [19{,}000,\ 23{,}000]$ and $\sigma \in [2{,}000,\ 5{,}500]$, 300 points each. Subtract the maximum before exponentiating (avoids underflow), divide by the total, then sum over σ (for μ) and over μ (for σ).



Left: joint posterior contours. The μ marginal is flat-topped/irregular because the Laplace likelihood is piecewise linear in μ (the posterior mode is the sample median).

› (b) Posterior means and fit.

$\mathrm{E}(\mu \mid \mathbf{Y}) = 20{,}856$ (sample median 20,834) and $\mathrm{E}(\sigma \mid \mathbf{Y}) = 3{,}091$.



Not a good fit, for the same reason as Exercise 9: the data have several clusters and the Laplace density has a single sharp peak.

› (c) Posterior predictive.

Draw 10^5 grid cells with probability equal to their posterior mass, then draw $Y^* \sim \mathrm{Laplace}(\mu, \sigma)$ for each.

	Mean	Variance
Plug-in Laplace(20,856, 3,091)	20,856	$2\hat{\sigma}^2 = 1.911 \times 10^7$
Posterior predictive	20,840	1.937×10^7

Why? By the laws of total expectation and variance: $\mathrm{E}(Y^*) = \mathrm{E}(\mu)$, so the means agree; $\mathrm{Var}(Y^*) = \mathrm{E}(2\sigma^2) + \mathrm{Var}(\mu) > 2[\mathrm{E}\sigma]^2$, so the PPD is wider — here by only about 1%, because with $n = 82$ the parameters are well determined.

Answer

(a) See the contour and marginal plots. (b) $\hat{\mu} \approx 20{,}856$, $\hat{\sigma} \approx 3{,}091$; the Laplace fits poorly (multimodal data). (c) Same mean; the PPD variance is about 1.3% larger than the plug-in variance, because it also accounts for uncertainty in (μ, σ).

Takeaway & common mistakes

The PPD is always at least as spread out as the plug-in distribution. The difference matters for small samples and disappears as n grows.

Exercise 3.11 — Reggie Jackson with MCMC diagnostics

Exercise 11 (exact text from the book)

In Section 2.5 we compared Reggie Jackson's home run rate in the regular season and World Series. He hit 563 home runs in 2820 regular-season games and 10 home runs in 27 World Series games (a player can hit 0, 1, 2, … home runs in a game). Assuming Uniform(0,10) priors for both home run rates, use JAGS to summarize the posterior distribution of (i) his home run rate in the regular season, (ii) his home run rate in the World Series, and (iii) the ratio of these rates. Provide trace plots for all three parameters and discuss convergence of the MCMC sampler including appropriate convergence diagnostics.

In plain words

Redo Exercise 2.2 with uniform priors and MCMC, and show how to check that the sampler has converged.

Key idea

A derived quantity (the ratio) is free in MCMC: compute it from each pair of draws. Check convergence with several chains from very different starting values.

Solution

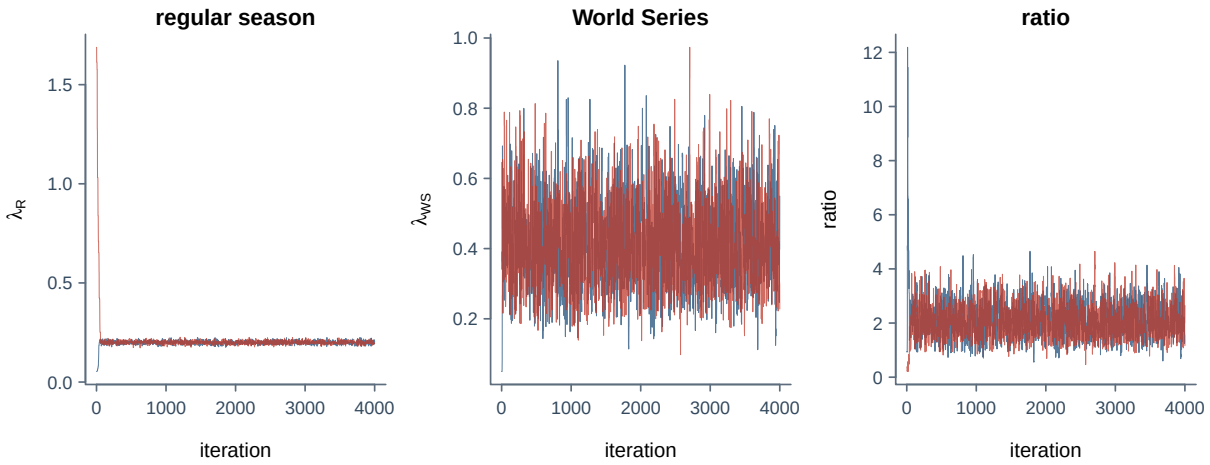
Model.

$Y_R \mid \lambda_R \sim \text{Poisson}(2820\lambda_R)$, $Y_{WS} \mid \lambda_{WS} \sim \text{Poisson}(27\lambda_{WS})$, $\lambda_R, \lambda_{WS} \overset{\text{iid}}{\sim} \text{Uniform}(0, 10)$. Exact posteriors: $\text{Gamma}(564, 2820)$ and $\text{Gamma}(11, 27)$ (truncated at 10, negligibly).

```
model{
  Y_reg ~ dpois(N_reg * lambda_reg);    Y_ws ~ dpois(N_ws * lambda_ws)
  lambda_reg ~ dunif(0, 10);            lambda_ws ~ dunif(0, 10)
  ratio ← lambda_ws / lambda_reg
}
```

Our own sampler.

Random-walk Metropolis on $\log \lambda$ (with the Jacobian term), two chains started at 0.05 and 2 HR/game, 20,000 iterations, 2,000 burn-in.



	Mean	SD	95% interval	$\hat{R}$	ESS	Exact mean
λ_R	0.200	0.008	(0.184, 0.217)	1.000	8,213	564/2820 = 0.200
λ_{WS}	0.409	0.123	(0.208, 0.682)	1.000	8,596	11/27 = 0.407
ratio	2.05	0.62	(1.04, 3.42)	1.000	8,607	—

$\text{Prob}(\lambda_{WS} > \lambda_R \mid \text{data}) = 0.98$.

› JAGS.

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	Mean	95% interval	$\hat{R}$	ESS
λ_R	0.200	(0.184, 0.217)	1.00	34,820
λ_{WS}	0.407	(0.205, 0.681)	1.00	29,250
ratio	2.04	(1.02, 3.43)	1.00	29,530

› Convergence discussion.

- **Trace plots**: both chains move quickly from their (deliberately bad) starts into the same region and overlap thereafter.
- **Gelman–Rubin**: $\hat{R} = 1.000$ for all three quantities (threshold 1.1).
- **ESS**: over 8,000 for each, far more than needed for means and 95% intervals.
- **Exact check**: MCMC means match the exact Gamma posterior means.

Answer

(i) λ_R: mean 0.200, 95% CI (0.184, 0.217). (ii) λ_{WS}: mean 0.41, 95% CI (0.21, 0.68). (iii) Ratio: mean 2.05, 95% CI (1.04, 3.42), $\text{Prob}(\text{ratio} > 1) = 0.98$. The sampler has clearly converged.

Takeaway & common mistakes

Compare with Exercise 2.2: the $\text{Uniform}(0, 10)$ prior (equivalent to $\text{Gamma}(1, 0)$, one "pseudo home run") pulls the World Series rate up from 0.37 to 0.41, making the ratio interval exclude 1. With only 10 home runs, the prior matters.

Exercise 3.12 — Is the discovery rate of bivalve species changing?

Exercise 12 (exact text from the book)

As discussed in Section 1.6, [47] report that the number of marine bivalve species discovered each year from 2010-2015 was 64, 13, 33, 18, 30 and 20. Denote Y_t as the number of species discovered in year $2009 + t$ (so that $Y_1 = 64$ is the count for 2010). Use JAGS to fit the model

$$Y_t | \alpha, \beta \overset{indep}{\sim} \text{Poisson}(\lambda_t)$$

where $\lambda_t = \exp(\alpha + \beta t)$ and $\alpha, \beta \sim \text{Normal}(0, 10^2)$. Summarize the posterior of α and β and verify that the MCMC sampler has converged. Does this analysis provide evidence that the rate of discovery is changing over time?

In plain words

Fit a Poisson regression with a log-linear time trend to six yearly counts and decide whether the trend slope β is non-zero.

Background you need

In $\lambda_t = e^{\alpha+\beta t}$, e^{β} is the *multiplicative* change in the expected count per year: $\beta = -0.18$ means $e^{-0.18} = 0.835$, a 16.5% decrease per year.

Key idea

The log posterior is easy to write: $\sum_t \left[Y_t(\alpha + \beta t) - e^{\alpha+\beta t}\right] + \log \pi(\alpha) + \log \pi(\beta)$. Neither full conditional is standard, so use Metropolis (Exercise 13) or JAGS.

Solution

JAGS model.

```
model{
  for(t in 1:6){
    Y[t] ~ dpois(lambda[t])
    log(lambda[t]) <- alpha + beta * t
  }
  alpha ~ dnorm(0, 0.01)        # precision 0.01 = variance 10^2
  beta  ~ dnorm(0, 0.01)
}
```

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	Mean	95% interval	$\hat{R}$	ESS
α	3.971	(3.666, 4.270)	1.001	5,071
β	−0.181	(−0.271, −0.094)	1.001	5,116

$\text{Prob}(\beta < 0 \mid \mathbf{Y}) = 0.99997.$

Posterior summary

(our Metropolis sampler from Exercise 13; three chains × 27,000 kept draws):

	Mean	SD	95% interval	$\hat{R}$	ESS
α	3.969	0.156	(3.657, 4.268)	1.000	1,994
β	-0.181	0.046	$(-0.270, -0.093)$	1.001	2,046

› Convergence.

Three chains from very different starting points ($(0,0)$, $(5,-1)$, $(2,1)$) overlap after a short burn-in (trace plots in Exercise 13); $\hat{R} \leq 1.001$ and ESS around 2,000 for both parameters.

› Is the rate changing?

$\text{Prob}(\beta < 0 \mid \mathbf{Y}) > 0.9999$ and the 95% interval for β excludes 0. The posterior median yearly change is $e^{-0.18} - 1 = -16.5\%$.

Sanity check

The classical Poisson GLM gives $\hat{\alpha} = 3.976$, $\hat{\beta} = -0.181$ — the same as the posterior means, as expected with vague priors.

Answer

α: mean 3.97 (3.66, 4.27); β: mean -0.181 $(-0.270, -0.093)$. Converged ($\hat{R} \approx 1$). Yes — strong evidence of a decreasing discovery rate, about 16% fewer discoveries per year.

Takeaway & common mistakes

Caution: the count of 64 in 2010 is a big outlier and the counts are over-dispersed (Chapter 1, Exercise 16). A Poisson model then gives over-confident intervals; a negative binomial model would be more honest and the evidence would be weaker.

Exercise 3.13 — Writing the Metropolis sampler by hand

Exercise 13 (exact text from the book)

Write your own Metropolis sampler (i.e., not JAGS) for the previous problem. Turn in commented code, report your candidate distribution for each parameters and corresponding acceptance ratio, and use trace plots for each parameter to show the chains have converged.

In plain words

Implement the sampler for Exercise 12 yourself and show it works.

Key idea

One-parameter-at-a-time random-walk Metropolis: propose a change to α only, accept/reject; then the same for β. Tune each step size separately.

Solution

Commented code

(from `ch03.R`):

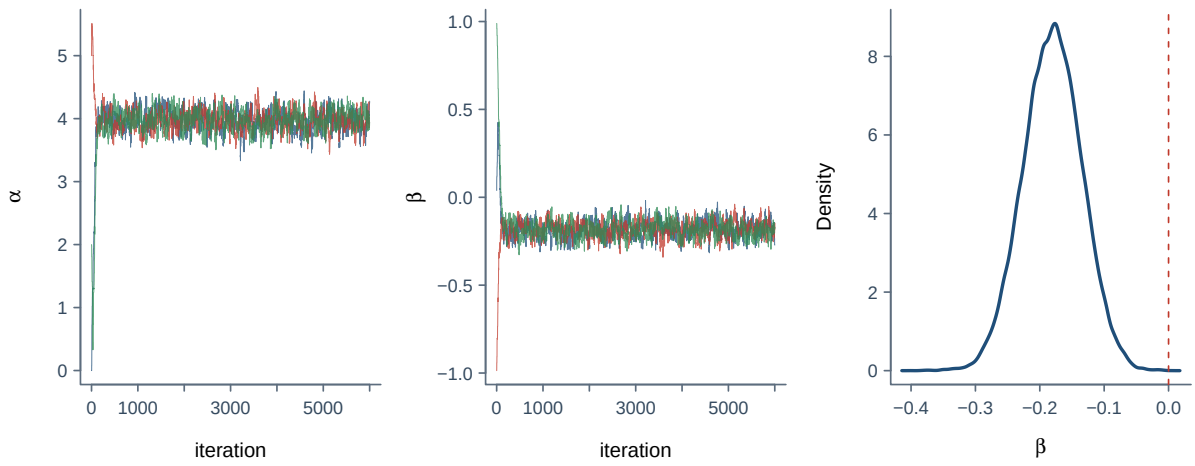
```
y ← c(64, 13, 33, 18, 30, 20); t ← 1:6
log_post ← function(ab) {                       # log posterior up to a constant
  eta ← ab[1] + ab[2] * t                       # log(lambda_t)
  sum(y * eta - exp(eta)) +                     # Poisson log-likelihood
    sum(dnorm(ab, 0, 10, log = TRUE))           # N(0, 10^2) priors
}
metro ← function(start, steps = c(0.12, 0.035), iters = 30000) {
  ab ← start; lp ← log_post(ab)
  for (it in seq_len(iters)) {
    for (j in 1:2) {                            # update alpha, then beta
      cand ← ab; cand[j] ← rnorm(1, ab[j], steps[j])   # normal random walk
      lp_cand ← log_post(cand)
      if (log(runif(1)) < lp_cand - lp) { ab ← cand; lp ← lp_cand }   # accept
    }
    draws[it, ] ← ab
  }
}
```

Candidate distributions and acceptance.

$\alpha^* \sim \text{Normal}(\alpha,\ 0.12^2)$ and $\beta^* \sim \text{Normal}(\beta,\ 0.035^2)$, chosen by trial and error. Acceptance rates: α 57%, β 57%.

Trace plots

(three chains, the first 6,000 iterations, plus the posterior of β):



Sanity check

$\hat{R} = 1.0004$ (α) and 1.0005 (β); the posterior means agree with JAGS and with the GLM.

Answer

Normal random-walk candidates with SD 0.12 (α) and 0.035 (β); acceptance $\approx 57\%$ each; trace plots and $\hat{R} \approx 1$ show convergence. Results as in Exercise 12.

Takeaway & common mistakes

The posterior correlation between α and β is -0.88, which slows a one-at-a-time sampler (ESS only $\approx 2{,}000$ out of 81,000). Centring time ($t - 3.5$ instead of t) would make α and β nearly uncorrelated and the sampler far more efficient.

Exercise 3.14 — Which dose is best?

Exercise 14 (exact text from the book)

A clinical trial assigned 100 patients each to the placebo, the low dose of a new drug, and the high dose of the new drug (for a total of 300 patients); the data are given in the table below.

Treatment	Positive outcome	Negative outcome
Placebo	52	48
Low dose	60	40
High dose	54	46

Conduct a Bayesian analysis using JAGS with uniform priors on the probability of a patient having a positive outcome under each treatment.

(a) Report the posterior mean and 95% interval for the probability of a positive outcome for each treatment group.

(b) Compute the posterior probability that the low dose is the best of the three treatment options.

In plain words

Three independent success rates; estimate each and the probability that the low dose has the highest rate.

Key idea

Each arm is a beta–binomial: $p_j \mid Y_j \sim \text{Beta}(1+Y_j,\ 1+100-Y_j)$. "Low dose is best" is the event $\{p_2 > p_1 \text{ and } p_2 > p_3\}$; its probability is the fraction of joint posterior draws in which it happens.

Solution

JAGS model.

```
model{
  for(j in 1:3){
    Y[j] ~ dbin(p[j], 100)
    p[j] ~ dunif(0, 1)
  }
  low_best <- step(p[2] - p[1]) * step(p[2] - p[3])   # 1 if low dose is best
}
```

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

Posterior means $p = (0.519,\ 0.598,\ 0.539)$; mean of `low_best` = Prob(low dose best) $= 0.739$; all $\hat{R} = 1.00$.

(a) Posteriors

(10^6 exact draws from each Beta posterior):

Arm	Posterior	Mean	95% interval
Placebo	$\text{Beta}(53, 49)$	0.520	(0.423, 0.615)
Low dose	$\text{Beta}(61, 41)$	0.598	(0.502, 0.691)
High dose	$\text{Beta}(55, 47)$	0.539	(0.442, 0.635)

(b) Probability each arm is best.

For each joint draw (p_1, p_2, p_3), record which is largest:

$$\text{Prob}(\text{placebo best}) = 0.093, \qquad \text{Prob}(\textbf{low dose best}) = \mathbf{0.737}, \qquad \text{Prob}(\text{high dose best}) = 0.170.$$

Answer

(a) Placebo 0.520 (0.42, 0.62); low dose 0.598 (0.50, 0.69); high dose 0.539 (0.44, 0.63). (b) $\text{Prob}(\text{low dose best} \mid \text{data}) \approx 0.74$.

Takeaway & common mistakes

The intervals overlap a lot, so 74% is suggestive but far from conclusive. The ranking probability is a quantity that is very natural in Bayesian analysis but awkward to obtain from classical tests.

Exercise 3.15 — A normal mixture with vague but proper priors

Exercise 15 (exact text from the book)

Consider the normal mixture model

$$Y_i|\theta \overset{\text{iid}}{\sim} f(y|\theta) = \frac{1}{2}[\phi(y-\theta) + \phi(y)],$$

for $i = 1, \ldots, n$ where $\theta \in \mathbb{R}$ and $\phi(z) = \exp\{-z^2/2\}/\sqrt{2\pi}$ denotes the density of a standard normal distribution. We have shown (Section 2.4, problem 14) that an improper prior cannot be used for this likelihood. In this problem we explore the use of vague but proper priors. Analyze data simulated in R as:

```
> set.seed(27695)
> theta_true <- 4
> n          <- 30
> B          <- rbinom(n,1,0.5)
> Y          <- rnorm(n,B*theta_true,1)
```

(a) Argue that the R code above generates samples from $f(y|\theta)$.

(b) Plot the simulated data, and plot $f(y|\theta)$ for $y \in [-3, 10]$ separately for $\theta = \{2, 4, 6\}$.

(c) Assuming prior $\theta \sim \text{Normal}(0, 10^2)$, obtain the MAP estimate of θ and its asymptotic posterior standard deviation using the following R code:

```
> nlp <- function(theta,Y){
>   like <- 0.5*dnorm(Y,0,1)+
>           0.5*dnorm(Y,theta,1)
>   prior <- dnorm(theta,0,10)
>   neg_log_post <- -sum(log(like))-sum(log(prior))
> return(neg_log_post)}
> map     <- optim(mean(Y),nlp,Y=Y,hessian=TRUE)
> map_est <- map$par
> map_sd  <- 1/sqrt(map$hessian)
```

(d) Suppose the prior distribution is $\theta \sim N(0, 10^k)$. Plot the posterior density of θ for $k \in \{0, 1, 2, 3\}$, and compare these posterior densities with the asymptotic normal distribution in (c) (by overlaying all five densities on the same plot).

(e) Use JAGS to fit this model via its mixture representation $Y_i|B_i, \theta \sim \text{Normal}(B_i\theta, 1)$, where $B_i \overset{\text{iid}}{\sim} \text{Bernoulli}(0.5)$ and $\theta \sim \text{Normal}(0, 10^2)$. Compare the posterior distribution of θ with the results from part (d).

In plain words

Half the data are $\text{N}(0, 1)$ noise, half are centred at the unknown θ. Estimate θ three ways — MAP + normal approximation, exact grid posterior for several priors, and a latent-variable Gibbs sampler — and compare.

Background you need

Law of total probability for densities: if $B \sim \text{Bernoulli}(0.5)$ and $Y \mid B \sim \text{N}(B\theta, 1)$, then $f(y) = 0.5\,\phi(y) + 0.5\,\phi(y-\theta)$.

Key idea

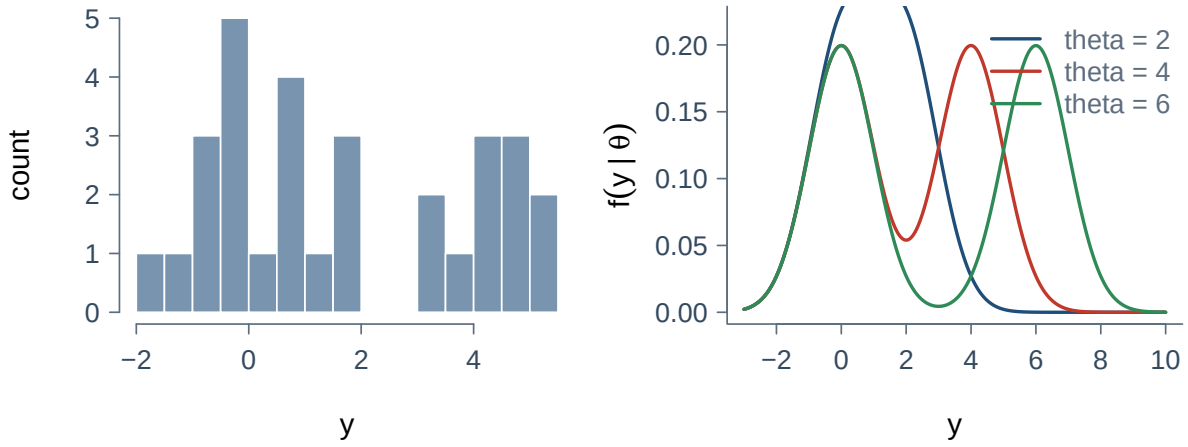
Adding the latent labels B_i makes the model easy: given the labels, θ is a normal mean estimated from the $B_i = 1$ observations only.

Solution

(a) Why the code samples from f.

For each i, `rbinom` flips a fair coin B_i; then `rnorm(n, B*theta, 1)` draws from $\mathrm{N}(0, 1)$ if $B_i = 0$ and from $\mathrm{N}(\theta, 1)$ if $B_i = 1$. Averaging over the coin (Recall box), the density of Y_i is $\frac{1}{2}\phi(y) + \frac{1}{2}\phi(y - \theta) = f(y \mid \theta)$.

(b) Data and densities.



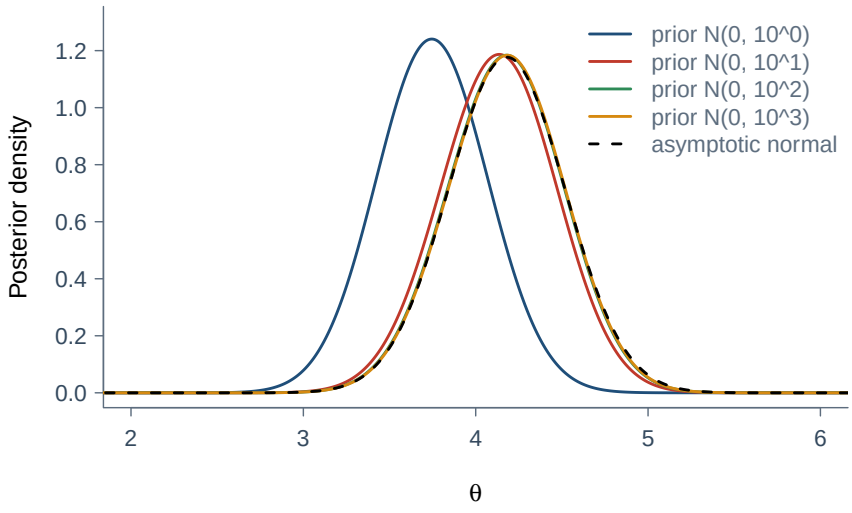
The data show two bumps (near 0 and near 4). For $\theta = 2$ the two components overlap into one lumpy bump; for $\theta = 4, 6$ they separate.

(c) MAP and asymptotic SD.

Running the given code: $\hat{\theta}_{MAP} = \mathbf{4.179}$ and asymptotic SD $= \mathbf{0.339}$, i.e. $\theta \mid \mathbf{Y} \approx \mathrm{N}(4.179,\ 0.339^2)$.

(d) Exact posteriors for four priors

(grid of 4,001 values; prior variance 10^k):



Prior	Posterior mean	Posterior SD
$\mathrm{N}(0, 10^0)$	3.749	0.320
$\mathrm{N}(0, 10^1)$	4.129	0.334
$\mathrm{N}(0, 10^2)$	4.171	0.335
$\mathrm{N}(0, 10^3)$	4.175	0.335
asymptotic normal (c)	4.179	0.339

Only the tight $\mathrm{N}(0, 1)$ prior (SD 1, which considers $\theta = 4$ very unlikely) pulls the posterior noticeably toward 0. For $k \geq 1$ all posteriors nearly coincide with each other and with the normal approximation.

› (e) Latent-variable sampler.

Gibbs updates: $B_i \mid \theta, Y_i \sim \text{Bernoulli}\left(\frac{\phi(Y_i-\theta)}{\phi(Y_i)+\phi(Y_i-\theta)}\right)$, and $\theta \mid \mathbf{B}, \mathbf{Y} \sim \mathrm{N}\left(\frac{\sum_{B_i=1} Y_i}{n_1+0.01}, \frac{1}{n_1+0.01}\right)$ where $n_1 = \sum B_i$. Our Gibbs sampler gives mean 4.177, SD 0.337 — matching the $k = 2$ grid posterior. JAGS:

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

θ: mean 4.169, SD 0.336, 95% interval (3.51, 4.82), $\hat{R} = 1.00$ — the same as the $k = 2$ curve in (d).

Answer

(a) Coin flip, then the chosen normal: the marginal is the mixture. (b) See plots. (c) MAP 4.179, SD 0.339. (d) For prior variance ≥ 10 the posterior is $\approx \mathrm{N}(4.17, 0.335^2)$; the $\mathrm{N}(0, 1)$ prior shrinks it to 3.75. (e) The latent-variable MCMC reproduces the $k = 2$ posterior.

Takeaway & common mistakes

A *proper* prior fixes the impropriety problem of Exercise 2.17, and here any reasonably vague proper prior gives the same answer — because the data do contain information about θ. The impropriety only bites in the far tails.

Exercise 3.16 — The shifted exponential distribution

Exercise 16 (exact text from the book)

Let $Y_1, \ldots, Y_n$ be a random sample from a shifted exponential distribution with density

$$f(y|\alpha,\beta) = \begin{cases} \beta \exp^{-\beta(y-\alpha)} & y \geq \alpha \\ 0 & y < \alpha \end{cases}$$

where $\alpha > 0$ and $\beta > 0$ are the parameters of interest. Assume prior distributions $\alpha \sim \text{Uniform}(0, c)$ and $\beta \sim \text{Gamma}(a, b)$.

(a) Plot $f(y|\alpha,\beta)$ for $\alpha = 2$ and $\beta = 3$ and $y \in [0, 5]$, give interpretations of α and β, and describe a real experiment where this model might be appropriate.

(b) Give the full conditional distributions of α and β, and determine if they are members of a common family of distributions.

(c) Write pseudo code for an MCMC sampler including initial values, full details of each sampling step, and how you plan to assess convergence.

In plain words

A waiting time that cannot be shorter than α and then decays exponentially at rate β. Interpret it, derive the Gibbs updates, and write the algorithm.

Background you need

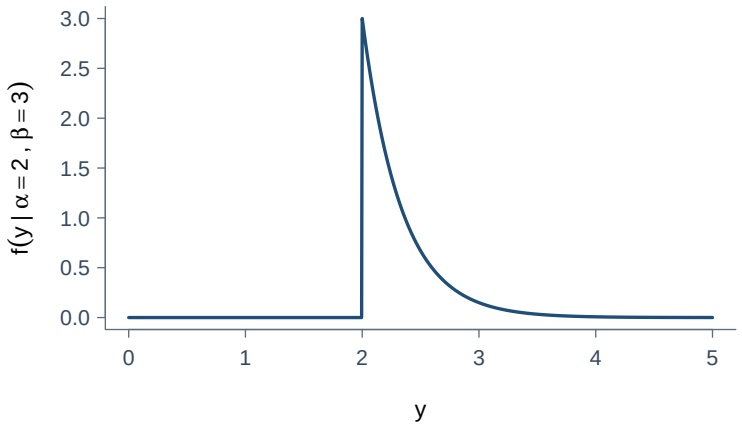
Indicator functions. The density is $\beta e^{-\beta(y-\alpha)}\mathbb{1}(y \geq \alpha)$. For a whole sample, $\prod_i \mathbb{1}(Y_i \geq \alpha) = \mathbb{1}(\alpha \leq \min_i Y_i)$: the threshold can be at most the smallest observation.

Key idea

The full conditional of α lives on a bounded interval, $0 < \alpha < \min(c, Y_{(1)})$, and is *increasing* there; the full conditional of β is a Gamma.

Solution

(a) Plot and interpretation.



α is a **minimum** (threshold, guarantee or lag time): no value can be smaller. β is the **rate** of the exponential part: the time beyond α has mean $1/\beta$ (here $\frac{1}{3}$). Example: the time until a customer's order is ready —

the kitchen needs at least α minutes, then random extra delay; or the lifetime of a component with a guaranteed minimum life.

› (b) Joint posterior.

$$p(\alpha, \beta \mid \mathbf{Y}) \propto \beta^n \exp\Big\{-\beta \sum_i (Y_i - \alpha)\Big\} \mathbb{1}(\alpha \le Y_{(1)}) \cdot \mathbb{1}(0 < \alpha < c) \cdot \beta^{a-1} e^{-b\beta}.$$

› (b) Full conditional of α.

Keep factors with α: $\exp\{-\beta \sum (Y_i - \alpha)\} = \exp\{-\beta \sum Y_i\}\, e^{n\beta\alpha}$. So

$$p(\alpha \mid \beta, \mathbf{Y}) \propto e^{n\beta\alpha} \quad \text{for } 0 < \alpha < u, \qquad u = \min(c,\, Y_{(1)}).$$

This is a *truncated exponential with a positive rate* (increasing density) — not a common named family, but it is easy to sample by inverting its CDF:

$$F(\alpha) = \frac{e^{n\beta\alpha} - 1}{e^{n\beta u} - 1} \;\Rightarrow\; \alpha = u + \frac{1}{n\beta} \log\big(U + (1-U)e^{-n\beta u}\big), \quad U \sim \text{Uniform}(0,1).$$

Why? Setting $F(\alpha) = U$ and solving gives $\alpha = \frac{1}{n\beta}\log\{1 + U(e^{n\beta u} - 1)\}$; we rewrite it in the form above (factoring out $e^{n\beta u}$) so that the computer never has to evaluate the enormous number $e^{n\beta u}$.

› (b) Full conditional of β.

$$p(\beta \mid \alpha, \mathbf{Y}) \propto \beta^{n+a-1} \exp\Big\{-\beta\Big(b + \sum_i (Y_i - \alpha)\Big)\Big\} \;\Rightarrow\; \beta \mid \alpha, \mathbf{Y} \sim \text{Gamma}\Big(a + n,\; b + \sum_i (Y_i - \alpha)\Big)$$

— a common family.

› (c) Pseudocode.

1. **Initial values**: $\alpha^{(0)} = \frac{1}{2}\min(c, Y_{(1)})$ (safely inside the allowed range), $\beta^{(0)} = 1/(\bar{Y} - \alpha^{(0)})$.
2. **For** $s = 1, \ldots, S$:
   (i) $U \sim \text{Uniform}(0,1)$, set $\alpha^{(s)} = u + \frac{1}{n\beta^{(s-1)}} \log\{U + (1-U)e^{-n\beta^{(s-1)}u}\}$;
   (ii) $\beta^{(s)} \sim \text{Gamma}\big(a + n,\; b + \sum_i (Y_i - \alpha^{(s)})\big)$.
3. **Convergence**: run 3 chains from dispersed starting values (e.g. $\alpha^{(0)}$ near 0, middle and near u), discard the first 1,000 draws, check trace plots, $\hat{R} < 1.01$ and ESS $> 1{,}000$ for both parameters.

Sanity check

We tested the sampler on 40 simulated values with $\alpha = 2$, $\beta = 3$ (priors $\text{Uniform}(0,5)$, $\text{Gamma}(1,1)$): posterior means $\alpha = 1.999$ (95% CI 1.978–2.007) and $\beta = 3.25$ (2.32–4.35). Both truths are recovered; α sits just below the smallest observation (2.007), exactly as the theory says.

Answer

(a) α = minimum value, β = decay rate (mean excess $1/\beta$). (b) $\alpha \mid \cdot \propto e^{n\beta\alpha}$ on $(0, \min(c, Y_{(1)}))$ (truncated exponential, not a standard family but invertible); $\beta \mid \cdot \sim \text{Gamma}(a+n,\; b + \sum(Y_i - \alpha))$. (c) Gibbs sampler above.

Takeaway & common mistakes

The posterior of a threshold parameter is very concentrated (its uncertainty shrinks like $1/n$, not $1/\sqrt{n}$) and piles up against the smallest observation.

Exercise 3.17 — A nonlinear growth curve for *T. rex*

Exercise 17 (exact text from the book)

Using the data (mass and age) provided in Listing 3.9, fit the following non-linear regression model:

$$\text{mass}_i \sim \text{Normal}(\mu_i, \sigma^2) \text{ where } \mu_i = \theta_1 + \theta_2 \text{age}_i^{\theta_3},$$

with priors $\theta_1 \sim \text{Normal}(0, 100^2)$, $\theta_2 \sim \text{Uniform}(0, 20000)$, $\theta_3 \sim \text{Normal}(0, 1)$, and $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$.

(a) Fit the model in JAGS and plot the data (age versus mass) along with the posterior mean of μ_i to verify the model fits reasonably well.

(b) Conduct a thorough investigation of convergence.

(c) Give three steps you might take to improve convergence.

In plain words

Fit a power-law growth curve to 7 dinosaur fossils and investigate how well the MCMC sampler behaves.

Background you need

Data (Listing 3.9): mass $= (29.9, 1761, 1807, 2984, 3230, 5040, 5654)$ kg at ages $(2, 15, 14, 16, 18, 22, 28)$ years.

Key idea

For a *fixed* θ_3, the model is linear in θ_1, θ_2, so they (and σ^2) can be Gibbs-sampled; only θ_3 needs Metropolis. But θ_2 and θ_3 trade off against each other (a larger exponent needs a smaller multiplier), which slows mixing.

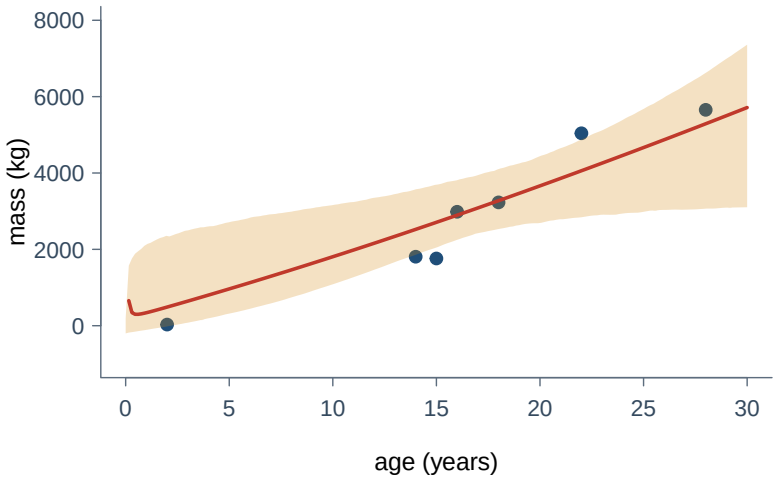
Solution

(a) Model and fit.

JAGS code:

```
model{
  for(i in 1:n){
    mass[i] ~ dnorm(mu[i], tau)
    mu[i] <- theta1 + theta2 * pow(age[i], theta3)
  }
  theta1 ~ dnorm(0, 1/100^2);   theta2 ~ dunif(0, 20000)
  theta3 ~ dnorm(0, 1);         tau ~ dgamma(0.01, 0.01)
}
```

Our sampler: Gibbs for θ_1 (normal), θ_2 (normal truncated to $(0, 20000)$), σ^2 (inverse-gamma), random-walk Metropolis for θ_3 (step 0.08, acceptance 40%). Four chains $\times$ 40,000 iterations, 5,000 burn-in.



The posterior mean curve (red, with 95% band) passes through the data: the fit is reasonable.

	Mean	SD	95% interval
θ_1	-16	99	$(-210,\ 179)$
θ_2	354	825	$(24,\ 2{,}139)$
θ_3	1.06	0.41	$(0.12,\ 1.67)$
σ	857	544	$(360,\ 2{,}298)$

JAGS:

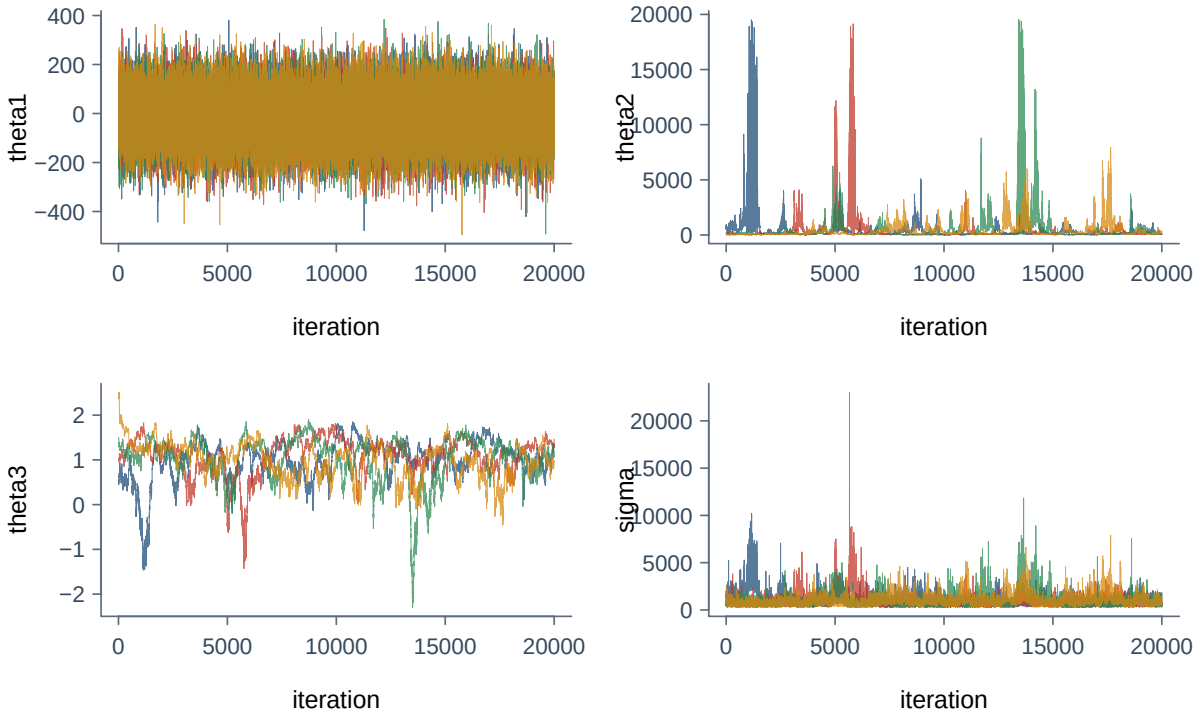
Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	Mean	95% interval	$\hat{R}$	ESS (of 150,000)
θ_1	-15.7	$(-211,\ 179)$	1.000	88,330
θ_2	235	(24, 1,183)	**1.153**	**159**
θ_3	1.10	(0.35, 1.66)	1.027	228
σ	795	(360, 1,867)	1.022	973

JAGS has the same trouble: $\hat{R} = 1.15$ for θ_2 signals that the chains have *not* converged, and the mean of θ_2 (235 vs. our 354) is unstable because of its long right tail.

› (b) Convergence investigation.



	$\hat{R}$	ESS (of 140,000)	lag-1 autocorrelation
θ_1	1.000	75,603	0.13
θ_2	1.006	370	0.88
θ_3	1.009	167	0.997
σ	1.005	410	0.56

θ_1 mixes well; θ_2 and θ_3 mix very slowly (lag-1 autocorrelation 0.997 means each draw is almost a copy of the previous one), and the trace plots of θ_2 show long excursions to large values. The chains agree ($\hat{R} < 1.01$), but the ESS of 167 for θ_3 from 140,000 draws is low: means are OK, tail quantiles are only roughly estimated. The cause is the strong dependence between θ_2 and θ_3 (correlation -0.72, and strongly nonlinear: the posterior is a curved "banana").

› (c) Three ways to improve convergence.

1. **Reparameterise / centre**: write $\mu_i = \theta_1 + \theta_2^*(\text{age}_i/20)^{\theta_3}$. Then θ_2^* is the mass at age 20, which is well determined whatever θ_3 is, so θ_2^* and θ_3 become nearly independent.
2. **Block updates**: propose (θ_2, θ_3) jointly, e.g. a bivariate normal random walk with the posterior covariance estimated from a pilot run (adaptive Metropolis).
3. **Run longer and/or thin**, and use several chains with good starting values (e.g. from a nonlinear least-squares fit) to reach the bulk of the posterior quickly.

Answer

(a) The posterior mean curve fits the 7 points well ($\theta_3 \approx 1.06$: roughly linear growth over this age range). (b) $\hat{R} < 1.01$ but θ_2, θ_3 mix very slowly (ESS 370 and 167, lag-1 autocorrelation 0.88 and 0.997) because they are strongly dependent. (c) Reparameterise (centre age), block-update (θ_2, θ_3), run longer chains with good inits.

Takeaway & common mistakes

$\hat{R} \approx 1$ is necessary but not sufficient: always look at ESS and autocorrelation too. Seven data points cannot pin down three curve parameters plus a variance — slow mixing is often a symptom of weak identification.

Exercise 3.18 — A binomial with unknown number of trials

Exercise 18 (exact text from the book)

Assume the (overly complicated) model $Y|n,p \sim \text{Binomial}(n,p)$ with prior distributions $n \sim \text{Poisson}(\lambda)$ and $p \sim \text{Beta}(a,b)$. The observed data is $Y = 10$.

(a) Describe why convergence may be slow for this model.

(b) Fit the model in JAGS with $\lambda = 10$ and $a = b = 1$. Check convergence and summarize the posterior of n, p and $\theta = np$.

(c) Repeat the analysis in (b) except with $a = b = 10$. Comment on the effect of the prior distribution of p on convergence for all the three parameters.

In plain words

We see 10 successes but know neither the number of trials n nor the success probability p. Only their product is really learned from the data. Show what that does to MCMC.

Background you need

From Chapter 1, Exercise 19: if $n \sim \text{Poisson}(\lambda)$ and $Y \mid n \sim \text{Binomial}(n,p)$, then $n - Y \mid Y,p \sim \text{Poisson}(\lambda(1-p))$. And $p \mid n, Y \sim \text{Beta}(a+Y,\ b+n-Y)$.

Key idea

The data identify $\theta = np$ (the expected count), not n and p separately. The posterior is a ridge along $np \approx 10$: large n with small p, or small n with large p. A Gibbs sampler that moves one coordinate at a time zig-zags slowly along such a ridge.

Solution

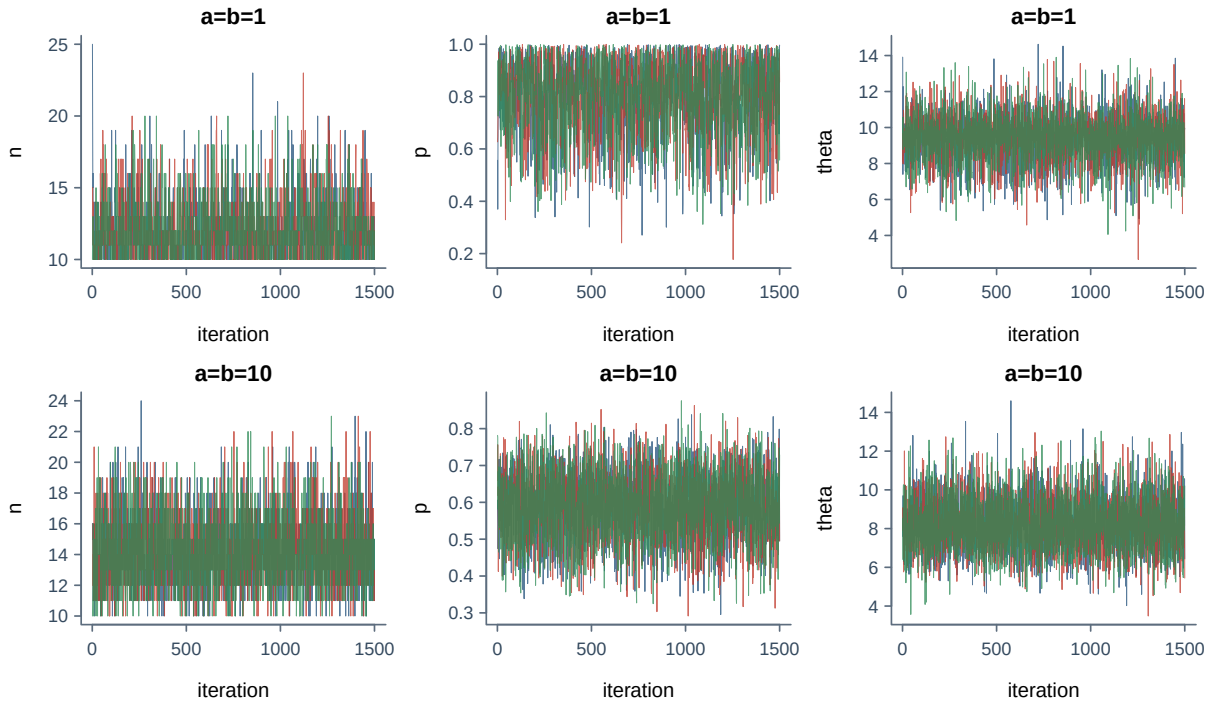
(a) Why convergence may be slow.

n and p are strongly negatively correlated in the posterior. Updating n given p (and then p given n) only allows small moves along the ridge, so successive draws are highly autocorrelated. Also n is discrete, so the sampler cannot make small adjustments.

(b), (c) Gibbs sampler and results.

Updates: $n = Y + M$ with $M \sim \text{Poisson}(\lambda(1-p))$, then $p \sim \text{Beta}(a+Y,\ b+n-Y)$. Three chains (starts $p = 0.05,\ 0.5,\ 0.95$), 20,000 iterations, 1,000 burn-in.

Prior		Mean	SD	95% interval	$\hat{R}$	ESS (of 57,000)	lag-1 acf
$a = b = 1$	n	11.99	1.99	(10, 17)	1.000	18,360	0.48
	p	0.801	0.141	(0.48, 0.99)	1.000	18,135	0.50
	θ	9.40	1.25	(6.6, 11.8)	1.000	57,000	−0.07
$a = b = 10$	n	14.13	2.22	(11, 19)	1.000	39,512	0.17
	p	0.589	0.091	(0.41, 0.76)	1.000	39,508	0.16
	θ	8.23	1.39	(5.7, 11.1)	1.000	57,000	−0.11



JAGS:

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

	$a=b=1$		$a=b=10$	
	mean	ESS	mean	ESS
n	12.03	19,710	14.12	42,530
p	0.798	18,770	0.588	42,280
θ	9.40	75,640	8.22	80,730

Same pattern: the informative prior on p more than doubles the ESS of n and p.

› Comments.

- With the flat prior ($a = b = 1$), n and p have lag-1 autocorrelation ≈ 0.5 and only about a third of the draws are "effective". With $\lambda = 10$ the prior already keeps n near 10–17, so the problem is moderate here; with a vaguer prior on n it would be much worse.
- With $a = b = 10$ the prior pins p near 0.5, which cuts the ridge: the autocorrelation drops to 0.17 and the ESS doubles for both n and p.
- $\theta = np$ mixes perfectly in both cases (ESS = number of draws): the identified combination is always easy; only the unidentified directions are slow.

Answer

(a) Only np is identified; the posterior ridge makes n and p highly correlated, so one-at-a-time updates move slowly. (b) $a = b = 1$: $n \approx 12$, $p \approx 0.80$, $\theta \approx 9.4$; converged but autocorrelated. (c) $a = b = 10$: $n \approx 14$, $p \approx 0.59$, $\theta \approx 8.2$; the informative prior on p roughly halves the autocorrelation of n and p, while θ mixes well in both cases.

⚠ Takeaway & common mistakes

When parameters are weakly identified, an informative prior is not only a statement of knowledge — it also makes computation easier. Report the identified quantity (θ) with confidence; be cautious about n and p separately.

Exercise 3.19 — An over-parameterised hierarchical model

Exercise 19 (exact text from the book)

Consider the (overly complicated) model

$$Y_i|\mu_i, \sigma_i^2 \sim \text{Normal}(\mu_i, \sigma_i^2)$$

where $\mu_i \overset{\text{iid}}{\sim} \text{Normal}(0, \theta_1^{-1})$, $\sigma_i^2 \sim \text{InvGamma}(\theta_2, \theta_3)$ and $\theta_j \sim \text{Gamma}(\epsilon, \epsilon)$ for $j = 1, 2, 3$. Assume the data are $Y_i = i$ for $i = 1, \ldots, n$.

(a) Describe why convergence may be slow for this model.

(b) Fit the model in JAGS with all four combinations of $n \in \{5, 25\}$ and $\epsilon \in \{0.1, 10\}$, and report the effective sample size for θ_1, θ_2 and θ_3 for all four fits.

(c) Comment on the effect of n and ϵ on convergence.

In plain words

Every observation gets its own mean *and* its own variance. With one data point per pair, the model has far more parameters than information. See how this affects MCMC efficiency.

Key idea

Each Y_i is a single number explained by two unknowns, μ_i and σ_i^2: a big Y_i can be "a big mean" or "a big variance". The hyperparameters θ_2, θ_3 are learned only indirectly through the barely-identified σ_i^2 — a recipe for slow mixing, especially with vague hyperpriors.

Solution

(a) Why convergence may be slow.

- **Weak identification**: μ_i and σ_i^2 are confounded (one observation each).
- θ_2 (shape) and θ_3 (scale) of the inverse-gamma are strongly correlated — many (shape, scale) pairs give similar distributions for the σ_i^2.
- With $\epsilon = 0.1$ the hyperpriors are vague and heavy-tailed, allowing the sampler to wander into regions such as $\theta_3 \to 0$ where it gets stuck.

Our sampler.

Gibbs for μ_i (normal), σ_i^2 (inverse-gamma), θ_1 ($\text{Gamma}(\epsilon + \frac{n}{2},\ \epsilon + \frac{1}{2}\sum \mu_i^2)$), θ_3 ($\text{Gamma}(\epsilon + n\theta_2,\ \epsilon + \sum \sigma_i^{-2})$), and Metropolis on $\log \theta_2$ (no conjugate form). 20,000 iterations, 2,000 burn-in, so the maximum ESS is 18,000.

(b) Effective sample sizes.

		ESS (of 18,000)			posterior median		
n	ϵ	θ_1	θ_2	θ_3	θ_1	θ_2	θ_3
5	0.1	1,054	168	146	0.089	1.57	0.001
5	10	7,532	2,550	11,641	0.874	0.81	1.10
25	0.1	12,513	**7**	**55**	0.004	1.83	0.049
25	10	4,164	3,206	10,558	0.915	0.29	1.49

JAGS:

Sanity check

JAGS output (run on GitHub Actions; 3 chains).

n	ϵ	ESS θ_1	ESS θ_2	ESS θ_3
5	0.1	983	144	338
5	10	8,777	6,355	13,260
25	0.1	7,967	**12**	104
25	10	5,202	10,691	15,018

(One chain of 20,000 draws each.) JAGS shows the same pattern as our sampler.

(c) Effect of n and ϵ.

- ϵ **matters most.** With $\epsilon = 10$ the hyperpriors are informative ($\text{Gamma}(10, 10)$: mean 1, SD 0.32), which keeps θ_2, θ_3 in a sensible region; every ESS is in the thousands.
- With vague hyperpriors ($\epsilon = 0.1$) the variance hyperparameters mix terribly: ESS of 7 for θ_2 when $n = 25$ — 20,000 iterations are worth about 7 independent draws. The posterior median of θ_3 near 0 shows the chain drifting toward a degenerate region.
- **More data does not automatically help.** Going from $n = 5$ to $n = 25$ improves θ_1 (the μ_i are informative about their common spread), but not θ_2, θ_3, because each new observation also brings two new unknowns.

Answer

(a) Each observation has its own mean and variance (confounded), and the inverse-gamma shape/scale hyperparameters are strongly dependent. (b) See the ESS table. (c) Informative hyperpriors ($\epsilon = 10$) give good mixing for all θ_j; vague ones ($\epsilon = 0.1$) give extremely poor mixing for θ_2, θ_3, and increasing n does not cure it.

Takeaway & common mistakes

If a model has (nearly) as many parameters as data points, expect MCMC trouble. Fixes: simplify the model (e.g. a common σ^2), use informative priors, or reparameterise.